UNIT-3 (DISCRETE MATHEMATICAL STRUCTURE)

GENERATING FUNCTION

LECTURE NO:-24

Generating Function

Generating Functions

Generating function is a method to solve the recurrence relations.

Let us consider, the sequence a_0, a_1, a_2....a_r of real numbers. For some interval of real numbers containing zero values at t is given, the function G(t) is defined by the series

$G(t)= a_0, a_1t+a_2 t^2+\cdots +a_r t_r+$.....equation (i)

This function G(t) is called the generating function of the sequence a_r.

Now, for the constant sequence 1, 1, 1, 1.....the generating function is

$$G(t) = \frac{1}{(1-t)}$$

It can be expressed as

$G(t) =(1-t)^{-1}=1+t+t^2 +t^3+t^4+\cdots$ [By binomial expansion]

Comparing, this with equation (i), we get

$a_0=1,a_1=1,a_2=1$ and so on.

For, the constant sequence 1,2,3,4,5,..the generating function is

$G(t) = \frac{1}{(1-t)^2}$ because it can be expressed as

$G(t) =(1-t)^{-2}=1+2t+3t^2 +4t^3+\cdots +(r+1) t^r$

Comparing, this with equation (i), we get
$a_0=1,a_1=2,a_2=3,a_3=4$ and so on.

The generating function of Z^r,($Z\neq0$ and Z is a constant)is given by

$G(t)= 1+Zt+Z^2 t^2+Z^3 t^3+\cdots +Z^r t^r$

$G(t)=\frac{1}{(1-Zt)}$ [Assume $|Zt|<1$]

So, $G(t)=\frac{1}{(1-Zt)}$ generates Z^r,$Z\neq0$

Also,If $a^{(1)}_r$ has the generating function $G_1(t)$ and $a^{(2)}_r$ has the generating function $G_2(t)$, then $\lambda_1 a^{(1)}_r+\lambda_2 a^{(2)}_r$ has the generating function $\lambda_1 G_1(t)+ \lambda_2 G_2(t)$. Here λ_1 and λ_2 are constants

Application Areas:

Generating functions can be used for the following purposes -

- For solving recurrence relations
- For proving some of the combinatorial identities
- For finding asymptotic formulae for terms of sequences

Example: Solve the recurrence relation $a_{r+2}-3a_{r+1}+2a_r=0$

By the method of generating functions with the initial conditions $a_0=2$ and $a_1=3$.

Solution: Let us assume that

$$G(t)=\sum_0^\infty a_r t^r$$

Multiply equation (i) by t^r and summing from r = 0 to ∞, we have

$$\sum_0^\infty a_{r+2} t^r - 3\sum_0^\infty a_{r+1} t^r + 2\sum_0^\infty a_r t^r=0$$

$(a_2+a_3 t+a_4 t^2+\cdots)-3(a_1+a_2 t+a_3 t^2+\cdots)+2(a_0+a_1 t+a_2 t^2+\cdots)=0$

$[\because G(t)=a_0+a_1 t+a_2 t^2+\cdots]$

$$\therefore \quad \frac{G(t)-a_0-a_1 t}{t^2} - 3\left(\frac{G(t)-a_0}{t}\right) +2G(t)=0 \text{.....equation (ii)}$$

Now, put $a_0=2$ and $a_1=3$ in equation (ii) and solving, we get

$$G(t) = \frac{2-3t}{1-3t+2t^2} \text{ or } G(t) = \frac{2-3t}{(1-t)(1-2t)}$$

$$\text{Now, Let } \frac{2-3t}{(1-t)(1-2t)} = \frac{A}{1-t} + \frac{B}{1-2t}$$

$$\text{i.e., } \quad 2-3t = A(1-2t)+B(1-t) \text{.....equation (iii)}$$

Put t=1 on both sides of equation (iii) to find A. Hence

$-1=-A \quad \therefore A = 1$

Put t=$\frac{1}{2}$ on both sides of equation (iii) to find B. Hence

$\frac{1}{2}=\frac{1}{2}$ B $\quad \therefore$ B = 1

Thus G (t) = $\frac{1}{1-t}+\frac{1}{1-2t}$.Hence,$a_r$=1+$2^r$.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

MITS

UNIT-4 (DISCRETE MATHEMATICAL STRUCTURE)

INTRODUCTION AND BASIC TERMINOLOGY OF GRAPH

LECTURE NO: -25

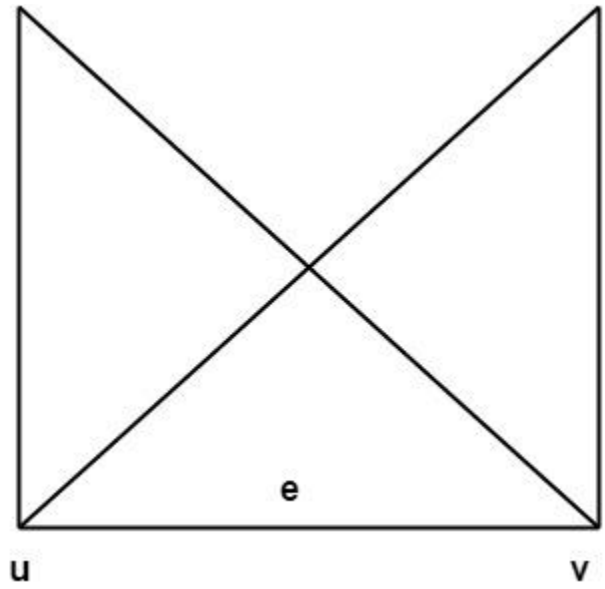
Introduction and Basic Terminology of Graph

Graph G consists of two things:

1. A set V=V(G) whose elements are called vertices, points or nodes of G.

2. A set E = E(G) of an unordered pair of distinct vertices called edges of G.

3. We denote such a graph by G(V, E) vertices u and v are said to be adjacent if there is an edge e ={u, v}.



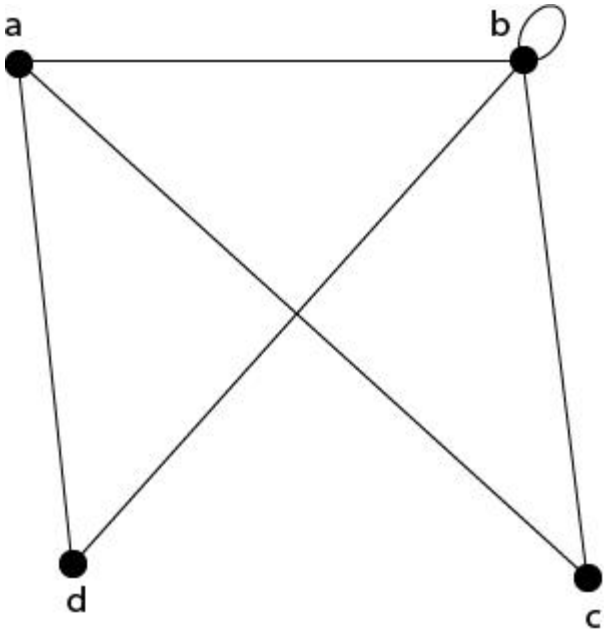
4. In such a case u and v are called the endpoint of e={u, v} and e are said to connect u and v.

Degree of a Vertex:

The degree of a vertex is the number of edges incident on a vertex v. The self-loop is counted twice. The degree of a vertex is denoted by d(v).

Example1: Consider the graph G shown in fig. Determine the degree of each vertex.

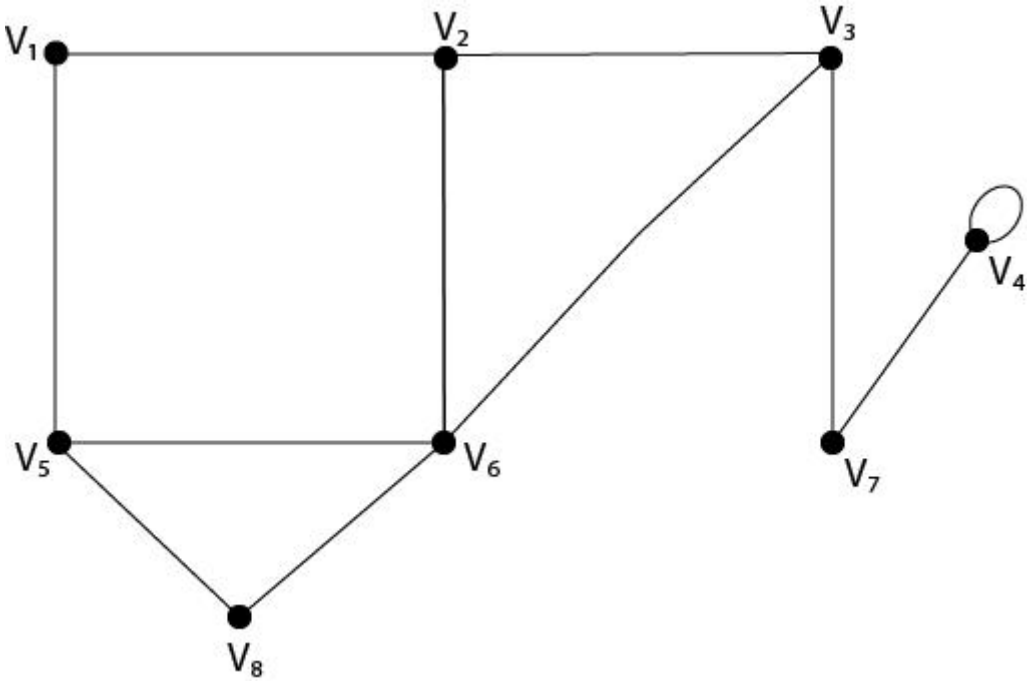
Notes By: -Sunil kumar Sharma (Assistant Professor-Maths Department)



Solution: The degree of each vertex is as follows:

d(a)=3; d(b)=5; d(c) = 2; d(d)=2.

Example2: Verify that the sum of the degrees of all the vertices is even for the graph shown in fig:

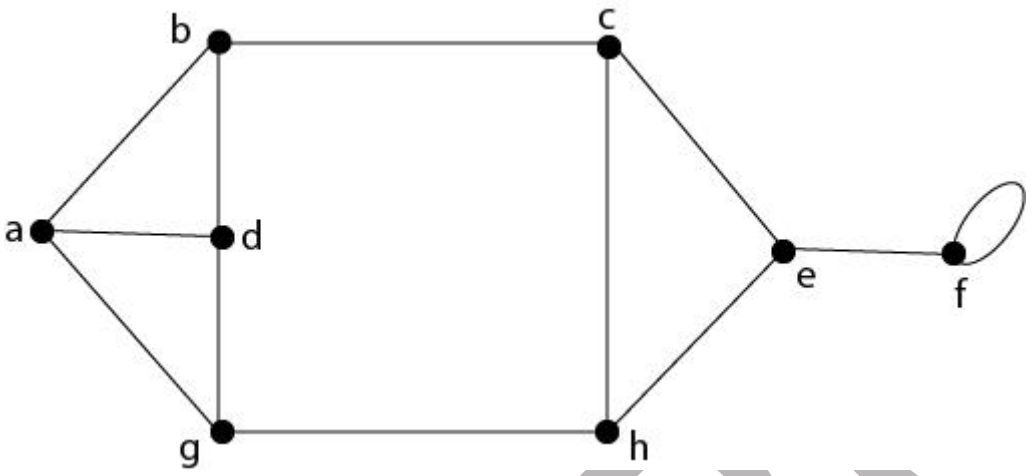


Solution: The sum of the degrees of all the vertices is:

$=d(v_1)+d(v_2)+d(v_3)+d(v_4)+d(v_5)+d(v_6)+d(v_7)+d(v_8)$

= 2+3+3+3+3+4+2+2=22, which is even.

Example3: Verify that there are even numbers of vertices of odd degrees in the graph shown in fig:



Solution: The number of vertices of degree odd is 8, and each has a degree three in the above graph. Hence, we have even number of vertices of odd degrees.

Pendant Vertex: A vertex with degree one is called a Pendant Vertex.

Pendant Edge: The only edge which is an incident with a pendant vertex is called the Pendant Edge.

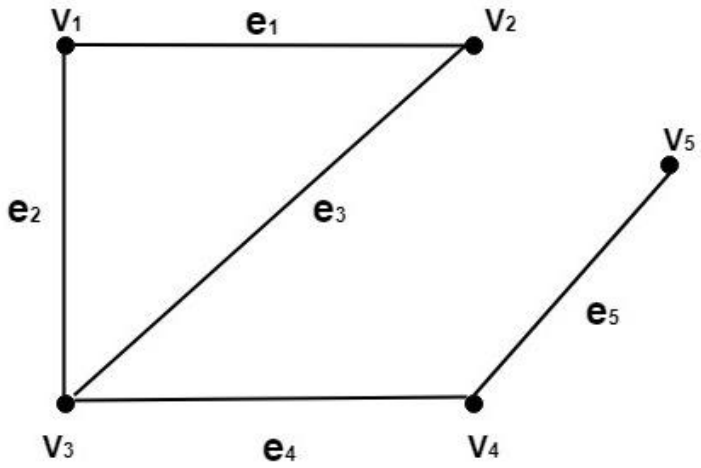
Odd Vertex: A vertex having degree odd is called an odd vertex.

Even Vertex: A vertex having a degree even is called an even vertex.

Incident Edge: An edge is called incident with the vertices is connects.

Adjacent Vertices: Two vertices are called adjacent if an edge links them. If there is an edge (u, v), then we can say vertex u is adjacent to vertex v, and vertex v is adjacent to vertex u.

Example: Consider the graph as shown in fig:



Determine the following:

1. Pendant Vertices
2. Pendant Edges
3. Odd vertices
4. Even Vertices
5. Incident Edges
6. Adjacent Vertices

Solution:

1. The vertex V_5 is the pendant vertex.
2. The edge (V_4,V_5) or e_5 is the pendant edge.
3. The vertices V_3 and V_5 are odd vertices.
4. The vertices V_1, V_2,and V_4 are even vertices.
5. The edge e_1 is an incident on V_1, and V_2.

   The edge e_2 is an incident on V_1 and V_3.

   The edge e_3 is an incident on V_2 and V_3.

   The edge e_4 is an incident on V_3 and V_4.

   The edge e_5 is an incident on V_4 and V_5.

   The vertex V_1 is adjacent to V_2 and V_3.

The vertex V_2 is adjacent to V_1 and V_2.

The vertex V_3 is adjacent to V_1 and V_4

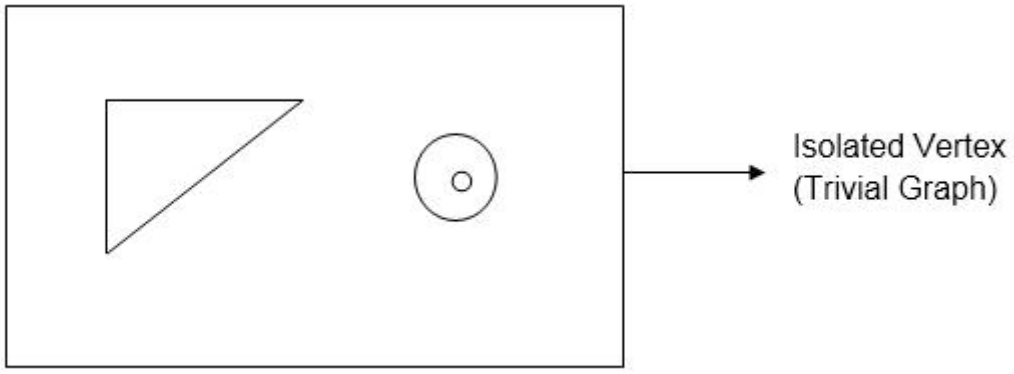
The vertex V_4 is adjacent to V_3 and V_5

The vertex V_5 is adjacent to V_4.

Self-Loop: A self-loop is an edge e if it has the same endpoint.

The graph shown in fig contains the self-loop at vertex b,i.e., e=(b, b).

Isolated Vertex: A vertex with degree 0 is called Isolated Vertex.



References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

UNIT-4 (DISCRETE MATHEMATICAL STRUCTURE)

TYPES OF GRAPHS

LECTURE NO: -26

Types of Graphs:

1. Null Graph: A null graph is defined as a graph which consists only the isolated vertices.

Example: The graph shown in fig is a null graph, and the vertices are isolated vertices.

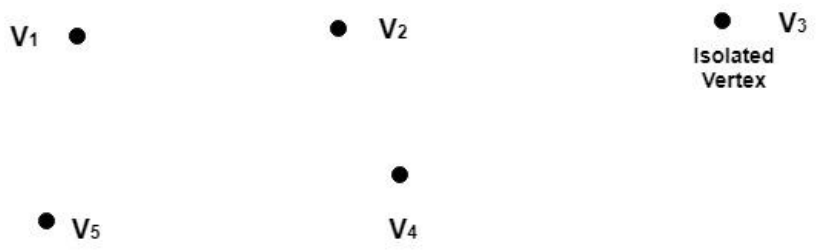


Fig:Null Graph

2. Undirected Graphs: An Undirected graph G consists of a set of vertices, V and a set of edge E. The edge set contains the unordered pair of vertices. If (u, v)∈E then we say u and v are connected by an edge where u and v are vertices in the set V.

Example: Let V = {1, 2, 3, 4} and E = {(1, 2), (1, 4), (3, 4), (2, 3)}.Draw the graph.

Solution: The graph can be drawn in several ways.

Two of which are as follows:

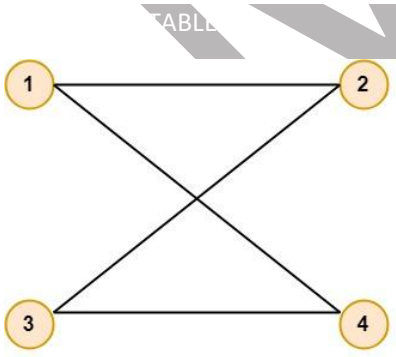


Fig:1

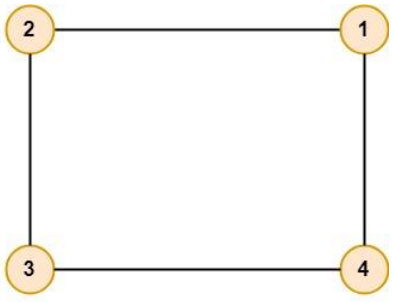


Fig:2

3. Multigraph: If in a graph multiple edges between the same set of vertices are allowed, it is known as Multigraph. In other words, it is a graph having at least one loop or multiple edges.

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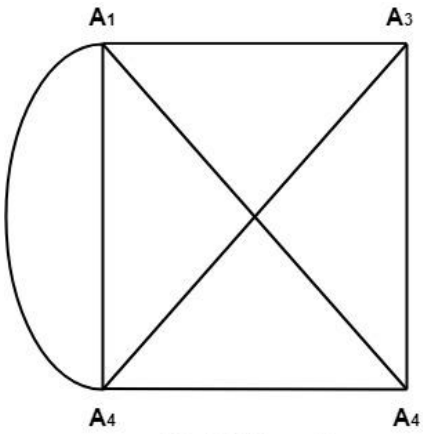
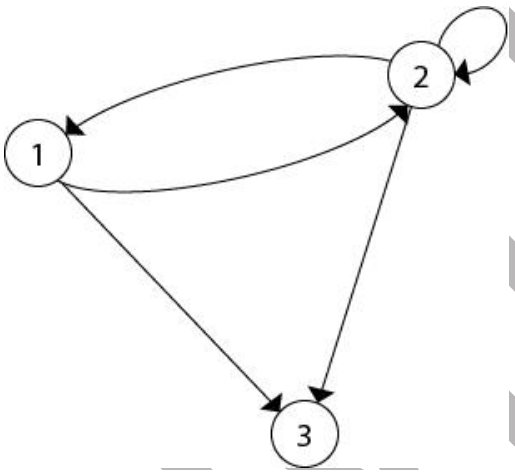


Fig:Multigraph

4. Directed Graphs: A directed graph or digraph G is defined as an unordered pair (V, E), where V is the set of points called vertices and E is the set of edges. Each edge in the graph G is assigned a direction and is identified with an ordered pair (u, v), where u is the initial vertex, and v is the end vertex.

Example: Consider the graph G = (V, E) as shown in fig. Determine the vertex set and edge set of graph G.



Solution: The vertex and edge set of graph G =(V, E) is as follow

G={{1,2,3},{(1,2),(2,1),(2,2),(2,3),(1,3)}}.

5. Undirected Complete Graph: An undirected complete graph G=(V,E) of n vertices is a graph in which each vertex is connected to every other vertex i.e., and edge exist between every pair of distinct vertices. It is denoted by K_n.A complete graph with n vertices will have $\frac{n(n-1)}{2}$ edges.

Example: Draw Undirected Complete Graphs $k_4$and k_6.

Solution: The undirected complete graph of k_4 is shown in fig1 and that of $k_6$is shown in fig2.

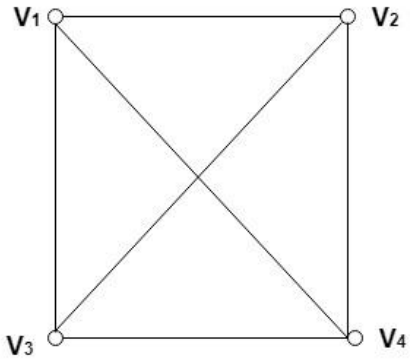


Fig:K_4

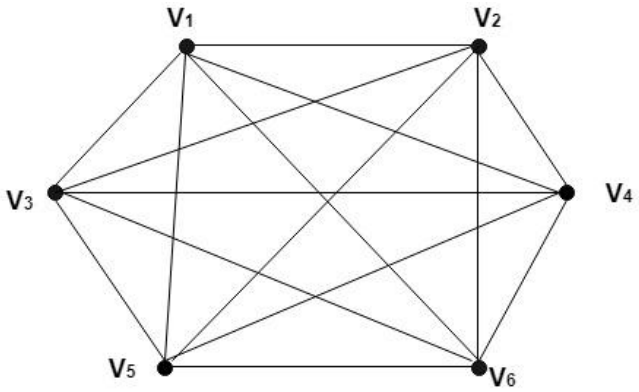


Fig:K_6

6. Connected and Disconnected Graph:

Connected Graph: A graph is called connected if there is a path from any vertex u to v or vice-versa.

Disconnected Graph: A graph is called disconnected if there is no path between any two of its vertices.

Example: Consider the graph shown in fig. Determine whether the graphs are

(a) Disconnected Graph

(b) Connected Graph.

Also, write their connected components.

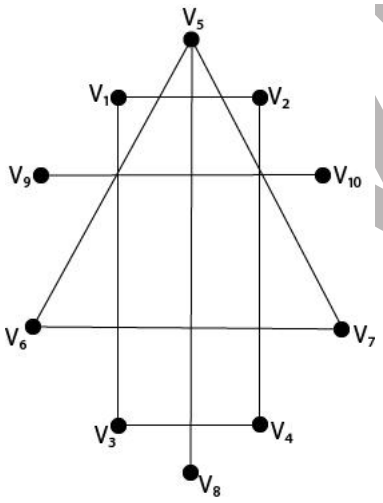


Fig:Disconnected Graph

Solution:

(i) The graph is shown in fig is a Disconnected Graph, and its connected components are

$\{V_1,V_2,V_3,V_4\}$,$\{V_5,V_6,V_7,V_8\}$ and $\{V_9,V_{10}\}$.

(ii) The graph shown in fig is a Disconnected Graph and its connected components are

$\{V_1,V_2\}$,$\{V_3,V_4\}$,$\{V_5,V_6\}$,$\{V_7,V_8\}$,$\{V_9,V_{10}\}$and $\{V_{11},V_{12}\}$.

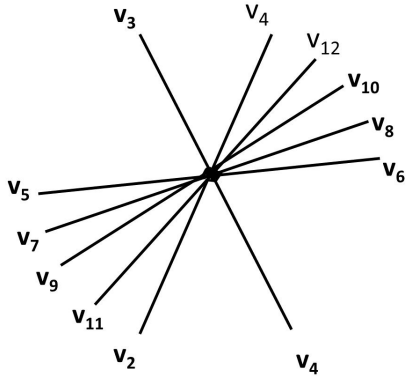


Fig: Disconnected graph

(iii) The graph shown in fig is a connected graph.

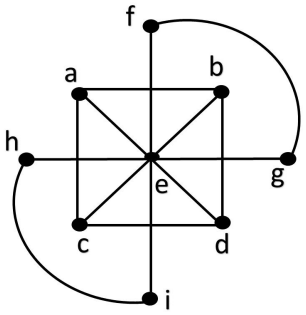
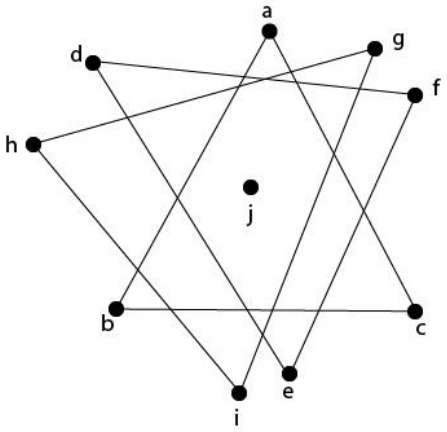


Fig: Connected graph

7. Connected Component: A subgraph of graph G is called the connected component of G, if it is not contained in any bigger subgraph of G, which is connected. It is defined by listing its vertices.

Example: Consider the graph shown in fig. Determine its connected components.



Solution: The connected components of this graph is {a, b, c}, {d, e, f}, {g, h ,i} and {j}.

8. Directed Complete Graph: A directed complete graph G = (V, E) on n vertices is a graph in which each vertex is connected to every other vertex by an arrow. It is denoted by K_n.

Example: Draw directed complete graphs K_3 and K_5.

Solution: Place the number of vertices at the appropriate place and then draw an arrow from each vertex to every other vertex as shown in fig:

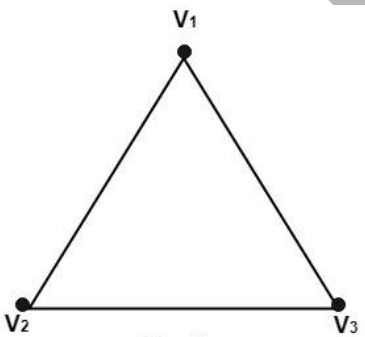


Fig:K_3

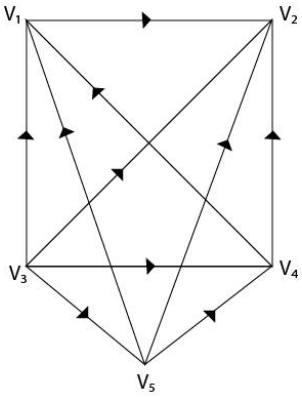
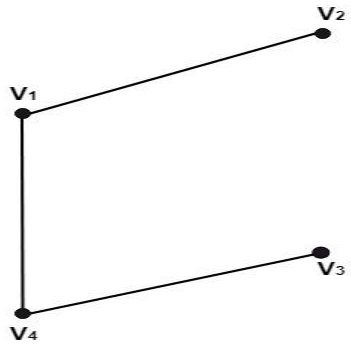


Fig: K_5

9. Complementary Graph: The complement of a graph G is defined to be a graph which has the same number of vertices as in graph G and has two vertices connected if and only they are not related in the graph.

Example: Consider the graph G shown in fig. Find the complement of this graph.



Solution: The complement of the above graph is shown in Fig:

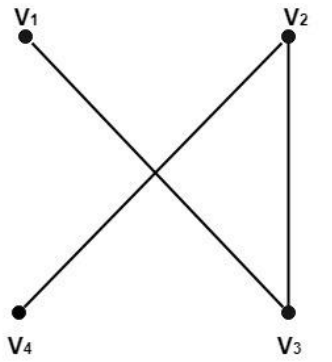


Fig:Complement of Graph G

10. Labeled Graphs: A graph G=(V, E) is called a labeled graph if its edges are labeled with some name or data. So, we can write these labels in place of an ordered pair in its edges set.

Example: The graph shown in fig is labeled graphs.

G= {{a, b, c, d}, {e_1,e_2,e_3,e_4}}

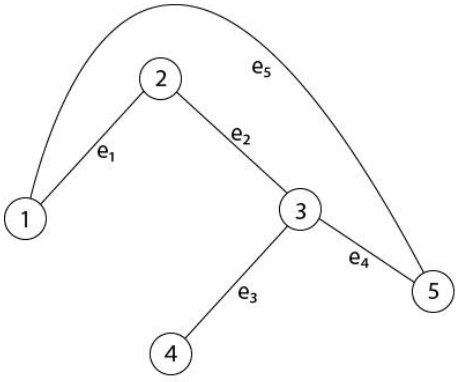


Fig: Undirected Labeled Graph

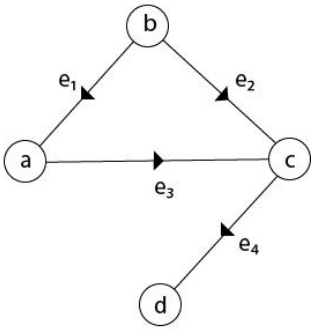


Fig: Directed Labeled Graph

11.Weighted Graphs: A graph G=(V, E) is called a weighted graph if each edge of graph G is assigned a positive number w called the weight of the edge e.

Example: The graph shown in fig is a Weighted Graph.

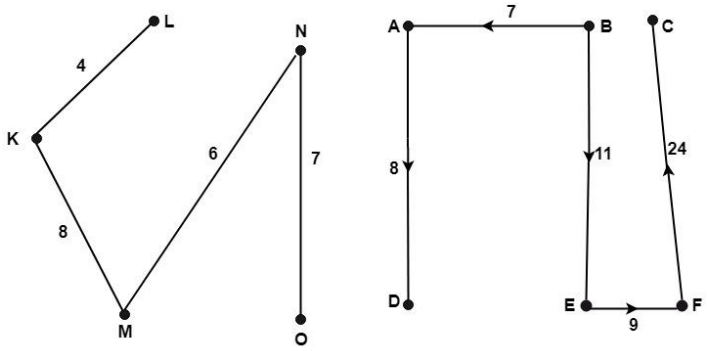
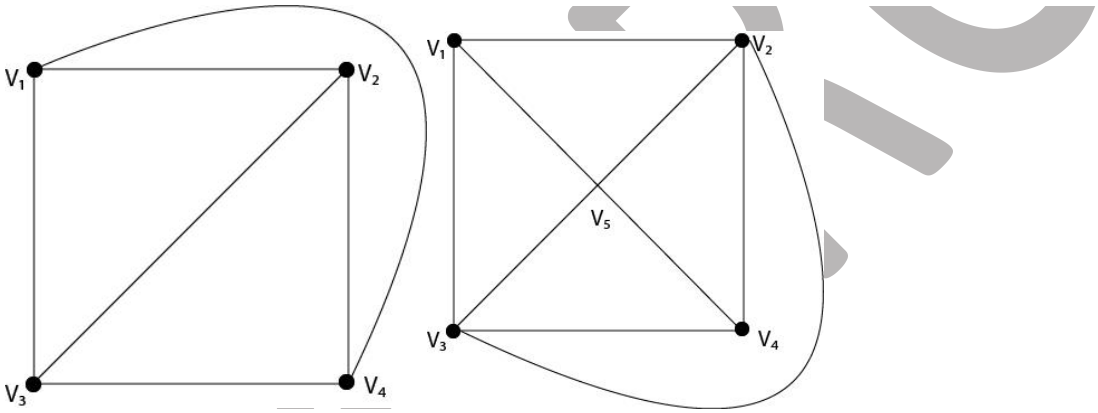


Fig:Weighted Graph

Planar Graph:

A graph is said to be planar if it can be drawn in a plane so that no edge cross.

Example: The graph shown in fig is planar graph.



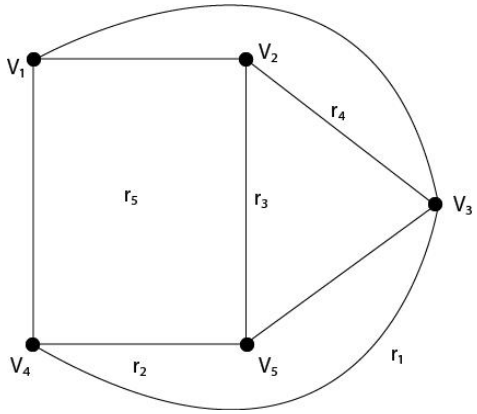
Region of a Graph: Consider a planar graph G=(V,E).A region is defined to be an area of the plane that is bounded by edges and cannot be further subdivided. A planar graph divides the plans into one or more regions. One of these regions will be infinite.

Finite Region: If the area of the region is finite, then that region is called a finite region.

Infinite Region: If the area of the region is infinite, that region is called a infinite region. A planar graph has only one infinite region.

Example: Consider the graph shown in Fig. Determine the number of regions, finite regions and an infinite region.

Example: Consider the graph shown in Fig. Determine the number of regions, finite regions and an infinite region.



Solution: There are five regions in the above graph, i.e. r_1,r_2,r_3,r_4,r_5.

There are four finite regions in the graph, i.e., r_2,r_3,r_4,r_5.

There is only one finite region, i.e., r_1

Properties of Planar Graphs:

1. If a connected planar graph G has e edges and r regions, then $r \leq \frac{2}{3} e$.
2. If a connected planar graph G has e edges, v vertices, and r regions, then v-e+r=2.
3. If a connected planar graph G has e edges and v vertices, then $3v-e \geq 6$.
4. A complete graph K_n is a planar if and only if $n<5$.
5. A complete bipartite graph K_{mn} is planar if and only if $m<3$ or $n>3$.

Example: Prove that complete graph K_4 is planar.

Solution: The complete graph K_4 contains 4 vertices and 6 edges.

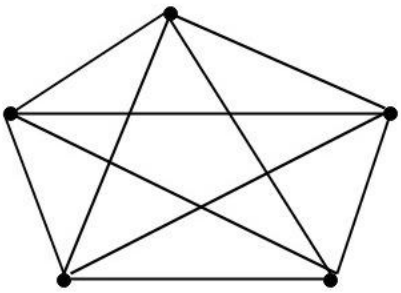
We know that for a connected planar graph $3v-e \geq 6$.Hence for K_4, we have 3x4-6=6 which satisfies the property (3).

Thus K_4 is a planar graph. Hence Proved.

Non-Planar Graph:

A graph is said to be non planar if it cannot be drawn in a plane so that no edge cross.

Example: The graphs shown in fig are non planar graphs.



K_5

These graphs cannot be drawn in a plane so that no edges cross hence they are non-planar graphs.

Properties of Non-Planar Graphs:

A graph is non-planar if and only if it contains a subgraph homeomorphic to K_5 or $K_{3,3}$

Example1: Show that K_5 is non-planar.

Solution: The complete graph K_5 contains 5 vertices and 10 edges.

Now, for a connected planar graph $3v-e\geq 6$.

Hence, for K_5, we have 3 x 5-10=5 (which does not satisfy property 3 because it must be greater than or equal to 6). Thus, K_5 is a non-planar graph.

Example2: Show that the graphs shown in fig are non-planar by finding a subgraph homeomorphic to K_5 or $K_{3,3}$.

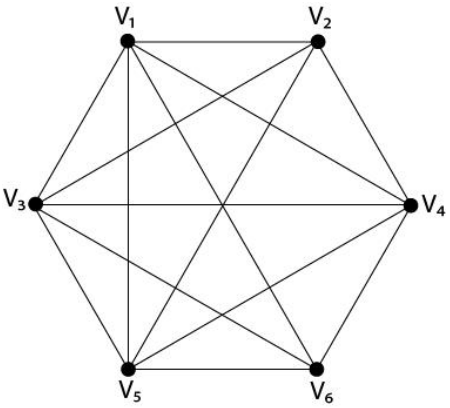


Fig: G_1

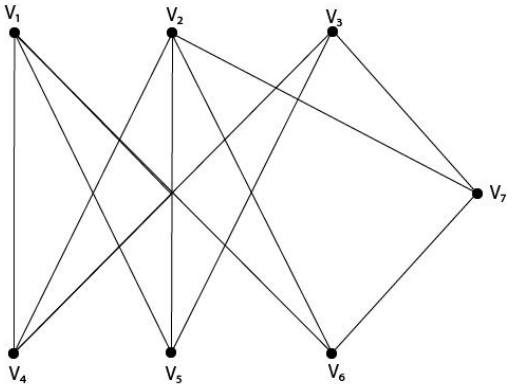


Fig: G_2

Solution: If we remove the edges (V_1,V_4),$(V_3,V_4)$and (V_5,V_4) the graph G_1,becomes homeomorphic to K_5.Hence it is non-planar.

If we remove the edge $V_{2,V}7$) the graph G_2 becomes homeomorphic to $K_{3,3}$.Hence it is a non-planar.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

MITRC

UNIT-4 (DISCRETE MATHEMATICAL STRUCTURE)

REPRESENTATION OF GRAPHS

LECTURE NO: -27

Representation of Graphs

There are two principal ways to represent a graph G with the matrix, i.e., adjacency matrix and incidence matrix representation.

(a)Representation of the Undirected Graph:

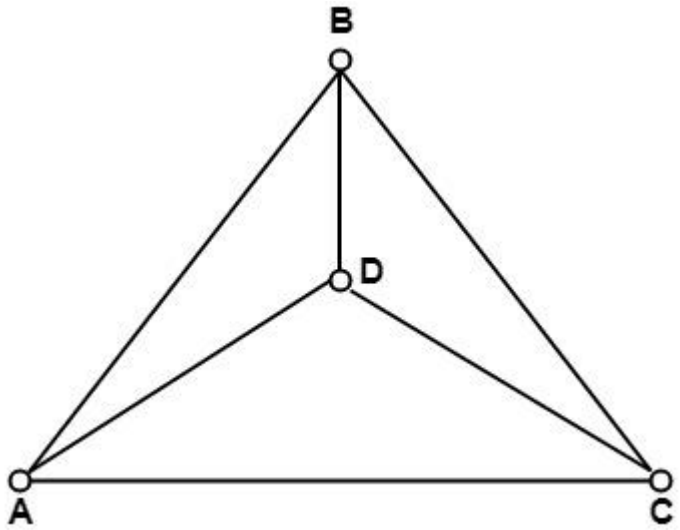
1. Adjacency Matrix Representation: If an Undirected Graph G consists of n vertices then the adjacency matrix of a graph is an n x n matrix A = $[a_{ij}]$ and defined by

$$a_{ij} = \begin{bmatrix} 1, \text{if } \{V_i, V_j\} \text{is an edge i.e., } v_i \text{is adjacent to } v_j \\ 0, \text{if there is no edge between } v_i \text{ and } v_j \end{bmatrix}$$

If there exists an edge between vertex v_i and v_j, where i is a row and j is a column then the value of a_{ij}=1.

If there is no edge between vertex v_i and v_j, then value of a_{ij}=0.

Example: Find the adjacency matrix M_A of graph G shown in Fig:



Solution: Since graph G consist of four vertices. Therefore, the adjacency matrix wills a 4 x 4 matrix. The adjacency matrix is as follows in fig:

Notes By: -Sunil kumar Sharma (Assistant Professor-Maths Department)

$$M_A = \begin{array}{c|cccc} & A & B & C & D \\ A & 0 & 1 & 1 & 1 \\ B & 1 & 0 & 1 & 1 \\ C & 1 & 1 & 0 & 1 \\ D & 1 & 1 & 1 & 0 \end{array}$$

2. Incidence Matrix Representation: If an Undirected Graph G consists of n vertices and m edges, then the incidence matrix is an n x m matrix C = [c_{ij}] and defined by

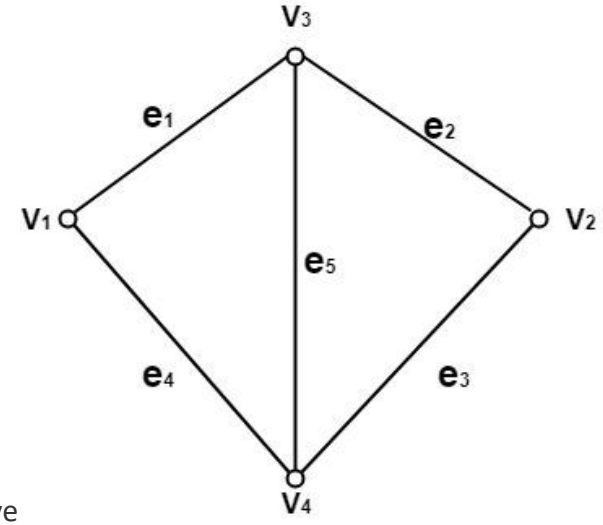
$$c_{ij} = \begin{bmatrix} 1, \text{if the vertex } V_i \text{incident by edge } e_j \\ 0, \text{otherwise} \end{bmatrix}$$

There is a row for every vertex and a column for every edge in the incident matrix.

The number of ones in an incidence matrix of the undirected graph (without loops) is equal to the sum of the degrees of all the vertices in a graph.

Example: Consider the undirected graph G as shown in fig. Find its incidence matrix M_I.

Solution: The undirected graph consists of four



ve rtices and five edges. Therefore, the incidence matrix is an 4 x 5 matrix, which is shown in Fig:

$$M_I = \begin{array}{c} \\ V_1 \\ V_2 \\ V_3 \\ V_4 \end{array} \begin{array}{c} e_1 \; e_2 \; e_3 \; e_4 \; e_5 \\ \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix} \end{array}$$

(b)Representation of Directed Graph:

1. Adjacency Matrix Representation: If a directed graph G consists of n vertices then the adjacency matrix of a graph is an n x n matrix A = $[a_{ij}]$ and defined by

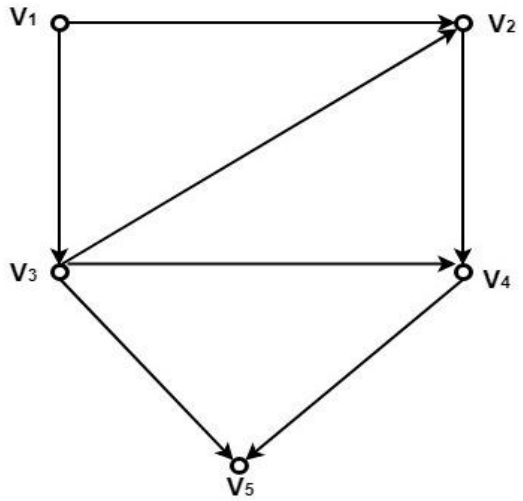
$$a_{ij} = \begin{bmatrix} 1, \text{if } \{V_i, V_j\} \text{is an edge i. e., } V_i \text{is initial vertex and } V_j \text{ is the final vertex} \\ 0, \text{if there is no edge between } V_i \text{ and } V_j \end{bmatrix}$$

If there exists an edge between vertex V_i and V_j, with V_i as initial vertex and V_j as a final vertex, then the value of a_{ij}=1.

If there is no edge between vertex V_i and V_j, then the value of a_{ij}=0.

The number of ones in the adjacency matrix of a directed graph is equal to the number of edges.

Example: Consider the directed graph shown in fig. Determine its adjacency matrix M_A.



Solution: Since the directed graph G consists of five vertices. Therefore, the adjacency matrix will be a 5 x 5 matrix. The adjacency matrix of the directed graphs is as follows:

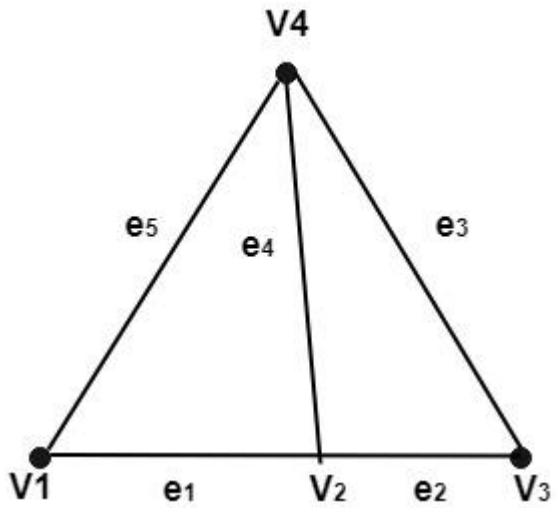
$$M_A = \begin{array}{c} \\ V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \end{array} \begin{array}{c} \begin{array}{ccccc} V_1 & V_2 & V_3 & V_4 & V_5 \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

2. Incidence Matrix Representation: If a directed graph G consists of n vertices and m edges, then the incidence matrix is an n x m matrix C = [c_{ij}] and defined by

$$c_{ij} = \begin{bmatrix} 1, \text{if } V_i \text{is the initial vertex of edge } e_j \\ -1, \text{if } V_i \text{is the final vertex of edge } e_j \\ 0, V_i \text{is not incident on edge } e_j \end{bmatrix}$$

The number of ones in an incidence matrix is equal to the number of edges in the graph.

Example: Consider the directed graph G as shown in fig. Find its incidence matrix M_I.



Solution: The directed graph consists of four vertices and five edges. Therefore, the incidence matrix is a 4 x 5 matrix which is show in fig:

$$|M_I = \begin{array}{c|ccccc} & e_1 & e_2 & e_3 & e_4 & e_5 \\ V_1 & -1 & 0 & 0 & 0 & -1 \\ V_2 & 1 & 1 & 0 & -1 & 0 \\ V_3 & 0 & -1 & 1 & 0 & 0 \\ V_4 & 0 & 0 & -1 & 1 & 1 \end{array}$$

(c)Representation of Multigraph:

Represented only by adjacency matrix representation.

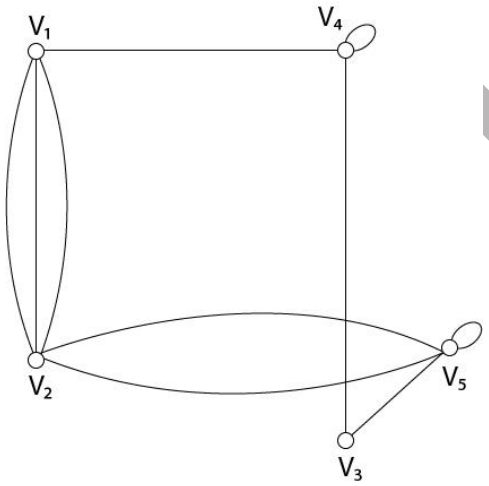
(i)Adjacency matrix representation of multigraph: If a multigraph G consists of vertices, then the adjacency matrix of graph is an n x n matrix A = $[a_{ij}]$ and is defined by

$$a_{ij} = \begin{bmatrix} N, \text{If there are more than one edges between vertex } v_i \text{and } v_j, \text{where} \\ N \text{ is the number of edges.} \\ 0, \text{Otherwise.} \end{bmatrix}$$

If there exist one or more than one edges between vertex v_i and v_j then a_{ij}=N, where is the number of edges between v_i and v_j.

If there is no edge between v_i and v_j.

Example: Consider the multigraph shown in Fig, Determine its adjacency matrix.



Solution: Since the multigraph consist of five vertices. Therefore the adjacency matrix will be an 5 x 5 matrix. The adjacency matrix of the multigraph is as follows:

$$M_A = \begin{array}{c|ccccc} & V_1 & V_2 & V_3 & V_4 & V_5 \\ V_1 & 0 & 3 & 0 & 0 & 1 \\ V_2 & 3 & 0 & 0 & 0 & 2 \\ V_3 & 0 & 0 & 0 & 1 & 1 \\ V_4 & 1 & 0 & 1 & 1 & 0 \\ V_5 & 0 & 2 & 1 & 0 & 1 \end{array}$$

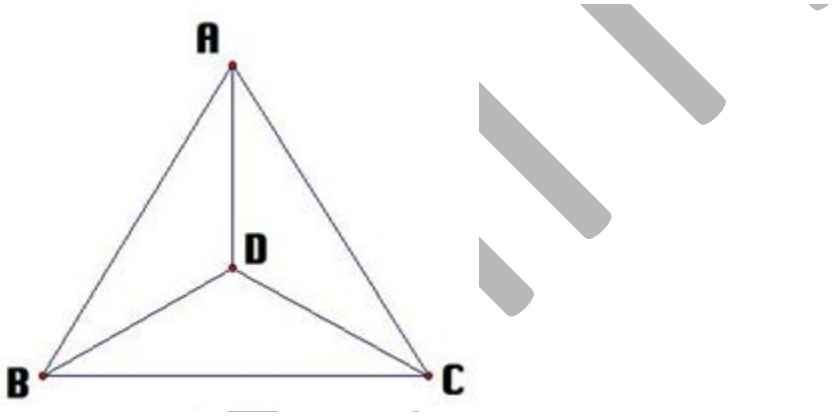
Basic Properties of Graph Theory

Properties of graph theory are basically used for characterization of graphs depending on the structures of the graph. Following are some basic properties of graph theory:

1 Distance between two vertices

Distance is basically the number of edges in a shortest path between vertex X and vertex Y. If there are many paths connecting two vertices, then the shortest path is considered as the distance between the two vertices. Distance between two vertices is denoted by **d(X, Y)**.

Example



Suppose, we want to find the distance between vertex B and D, then first of all we have to find the shortest path between vertex B and D.

There are many paths from vertex B to vertex D:

- B -> C -> A -> D, length = 3
- B -> D, length = 1(Shortest Path)
- B -> A -> D, length = 2
- B -> C -> D, length = 2

- B -> C -> A -> D, length = 3

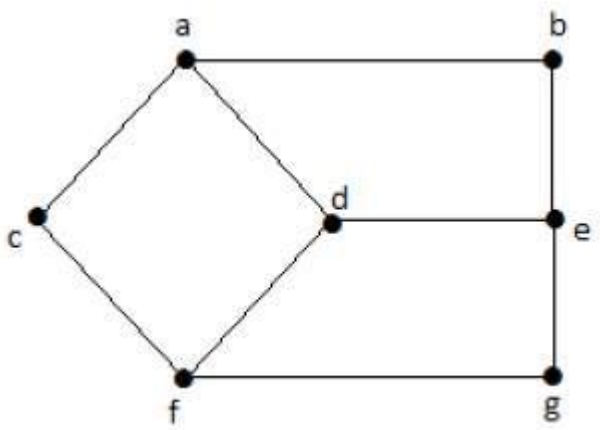
Hence, the minimum **distance between vertex B and vertex D is 1**.

2. Eccentricity of a vertex

Eccentricity of a vertex is the maximum distance between a vertex to all other vertices. It is denoted by **e(V)**.

To count the eccentricity of vertex, we have to find the distance from a vertex to all other vertices and **the highest distance is the eccentricity** of that particular vertex.

Example



In the above example, if we want to find the maximum eccentricity of vertex 'a' then:

- The distance from vertex a to b is 1 (i.e. ab)
- The distance from vertex a to c is 1 (i.e. ac)
- The distance from vertex a to f is 2 (i.e. ac -> cf or ad -> df)
- The distance from vertex a to d is 1 (i.e. ad)
- The distance from vertex a to e is 2 (i.e. ab -> be or ad -> de)
- The distance from vertex a to g is 3 (i.e. ab -> be -> eg or ac -> cf -> fg etc.)

Hence, the **maximum eccentricity of vertex** 'a' is 3, which is a maximum distance from vertex ?a? to all other vertices.

Similarly, maximum eccentricities of other vertices of the given graph are:

- e(b) = 3
- e(c) = 3
- e(d) = 2

- e(e) = 3
- e(f) = 3
- e(g) = 3

---

3. Radius of connected Graph

The **radius** of a connected graph is the **minimum eccentricity** from all the vertices. In other words, the minimum among all the distances between a vertex to all other vertices is called as the radius of the graph. It is denoted by **r(G)**.

Example

From the example of 5.2, r(G) = 2, which is the minimum eccentricity for the vertex 'd'.

4. Diameter of a Graph

Diameter of a graph is the **maximum eccentricity** from all the vertices. In other words, the maximum among all the distances between a vertex to all other vertices is considered as the diameter of the graph G. It is denoted by **d(G)**.

Example

From the above example, if we see all the eccentricities of the vertices in a graph, we will see that the diameter of the graph is the maximum of all those eccentricities.

Diameter of graph **d(G) = 3**, which is the maximum eccentricity.

---

5. Central point

If the eccentricity of the graph is equal to its radius, then it is known as **central point** of the graph.

Or

If **r(V) = e(V), then V is the central point** of the graph G.

Example

From the above example, 'd' is the central point of the graph. i.e.

1. e(d) = r(d) = 2

6. Centre

The set of all the central point of the graph is known as centre of the graph.

Example

From the example of 5.2, {**'d'**} is the centre of the graph.

---

7. Circumference

The total number of edges in the longest cycle of graph G is known as the circumference of G.

Example

In the above example, the **circumference is 6**, which is derived from the longest path a -> c -> f -> g -> e -> b -> a or a -> c -> f -> d -> e -> b -> a.

---

8. Girth

The total number of edges in the shortest cycle of graph G is known as **girth**. It is denoted by g(G).

Example

In the above example, the **girth of the graph is 4**, which is derived from the shortest cycle a -> c -> f -> d -> a, d -> f -> g -> e -> d or a -> b -> e -> d -> a.

9. Sum of degrees of vertices Theorem

For non-directed graph G = (V,E) where, Vertex set V = {V1, V2, ... Vn} then,

$$\sum_{i=1}^{n} \deg(V_i) = 2|E|$$

In other words, for any graph, the sum of degrees of vertices equals twice the number of edges.

Corollary 1

For directed graph G = (V, E) where, Vertex Set V = {V1, V2, ... Vn} then,

$$\sum_{i=1}^{n} \deg^{+}(V_i) = |E| = \sum_{i=1}^{n} \deg^{-}(V_i)$$

Corollary 2

The number of vertices in any non- directed graph with odd degree is even.

Example

It is impossible to make a graph with v (number of vertices) = 6 where the vertices have degrees 1, 2, 2, 3, 3, 4. This is because the sum of the degrees deg(V) is,

1. deg(V) = 1 + 2 + 2 + 3 + 3 + 4 = 15</p>
2. deg(V) is always an even number but 15 is odd!

Corollary 3

In an non-directed graph, if the degree of each vertex is k, then

1. $k|V| = 2|E|$

Corollary 4

If the degree of each vertex in a non-directed graph is at least k, then

1. $k|V| \leq 2|E|$

Corollary 5

If the degree of each vertex in a non- directed graph is at most k, then

1. $k|V| \geq 2|E|$

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

UNIT-4 (DISCRETE MATHEMATICAL STRUCTURE)

PATH, CIRCLE AND CONNECTIVITY

LECTURE NO: -28

Path, Circle and Connectivity

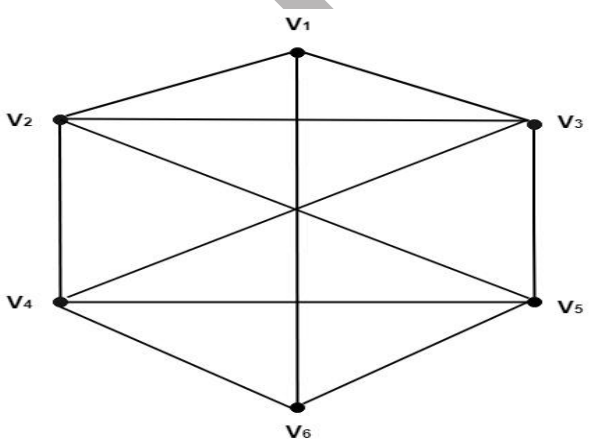
Path:

A path of length n is a sequence of n+1 vertices of a graph in which each pair of vertices is an edge of the graph.

1. **A Simple Path:** The path is called simple one if no edge is repeated in the path, i.e., all the vertices are distinct except that first vertex equal to the last vertex.
2. **An Elementary Path:** The path is called elementary one if no vertex is repeated in the path, i.e., all the vertices are distinct.
3. **Circuit or Closed Path:** The circuit or closed path is a path in which starts and ends at the same vertex, i.e., $v_0=v_n$.
4. **Simple Circuit Path:** The simple circuit is a simple path which is a circuit.

Example: Consider the graph shown in fig: Give an example of the following:

1. A simple path fromV_1 to V_6.
2. An elementary path from V_1 to V_6.
3. A simple path which is not elementary from V_1 to V_6.
4. A path which is not simple and starting fromV_2.
5. A simple circuit starting from V_1
6. A circuit which is not simple and starting from V_2.



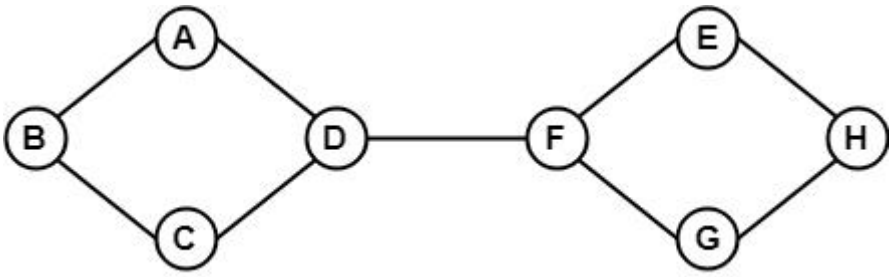
Solution:

1. A simple path from V_1 to V_6.
   $V_1, V_2, V_3, V_4, V_5, V_6$.
2. An elementary path from V_1 to V_6.
   $V_1, V_2, V_3, V_5, V_4, V_6$.
3. A simple path which is not elementary from V_1 to V_6.
   $V_1, V_2, V_3, V_5, V_2, V_4, V_6$.
4. A path which is not simple and starting from V_2.
   $V_2, V_3, V_4, V_5, V_3, V_4, V_6$.
5. A simple circuit starting from V_1.
   $V_1, V_2, V_4, V_6, V_5, V_3, V_1$
6. A circuit which is not simple and starting from V_2.
   $V_2, V_3, V_1, V_2, V_5, V_4, V_2$.

Connectivity

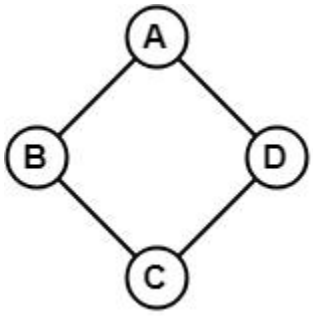
- **Connectivity** is a basic concept of graph theory. It defines whether a graph is connected or disconnected. Without connectivity, it is not possible to traverse a graph from one vertex to another vertex.
- A graph is said to be **connected graph if there is a path between every pair of vertex**. From every vertex to any other vertex there must be some path to traverse. This is called the connectivity of a graph.
- A graph is said to be **disconnected, if there exists multiple disconnected vertices and edges**.
- Graph connectivity theories are essential in network applications, routing transportation networks, network tolerance etc.

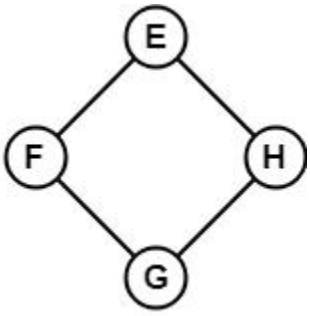
Example



In the above example, it is possible to travel from one vertex to another vertex. Here, we can traverse from vertex B to H using the path B -> A -> D -> F -> E -> H. Hence it is a connected graph.

Example





In the above example, it is not possible to traverse from vertex B to H because there is no path between them directly or indirectly. Hence, it is a disconnected graph.

Let's see some basic concepts of Connectivity.

1. Cut Vertex

A single vertex whose removal disconnects a graph is called a cut-vertex.

Let G be a connected graph. A vertex v of G is called a cut vertex of G, if G-v (Remove v from G) results a disconnected graph.

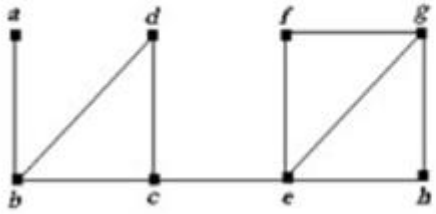
When we remove a vertex from a graph then graph will break into two or more graphs. This vertex is called a cut vertex.

Note: Let G be a graph with n vertices:

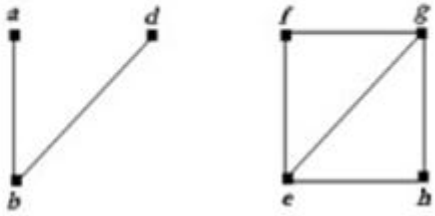
- A connected graph G may have maximum (n-2) cut vertices.
- Removing a cut vertex may leave a graph disconnected.
- Removing a vertex may increase the number of components in a graph by at least one.
- Every non-pendant vertex of a tree is a cut vertex.

Example 1

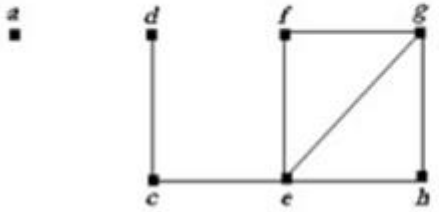
Original graph:



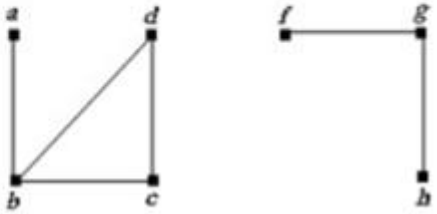
Vertex c is a cut vertex:



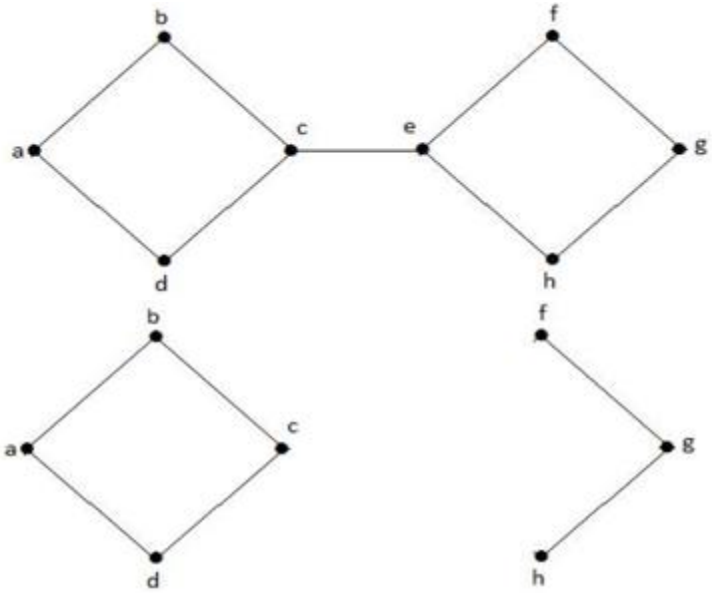
Vertex b is a cut vertex:



Vertex e is a cut vertex:



Example 2



In the above graph, vertex 'e' is a cut-vertex. After removing vertex 'e' from the above graph the graph will become a disconnected graph.

---

2. Cut Edge (Bridge)

A **cut- Edge or bridge** is a single edge whose removal disconnects a graph.

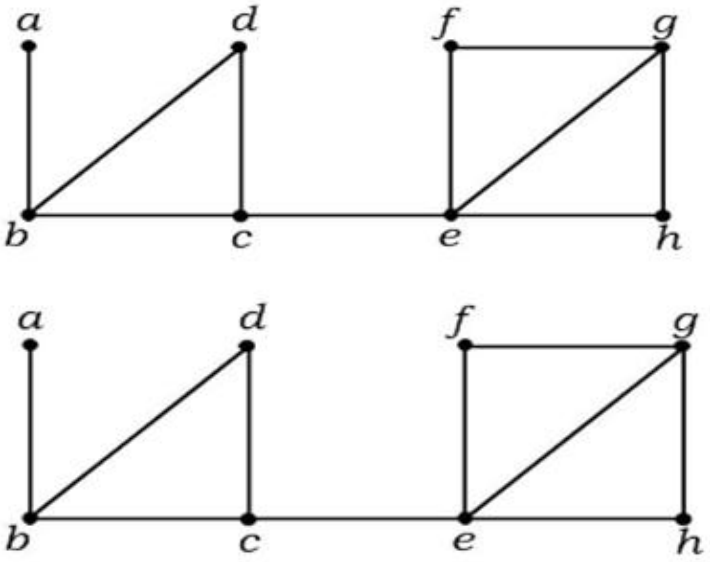
Let G be a connected graph. An edge e of G is called a cut edge of G, if G-e (Remove e from G) results a disconnected graph.

When we remove an edge from a graph then graph will break into two or more graphs. This removal edge is called a cut edge or bridge.

Note: Let G be a graph with n vertices:

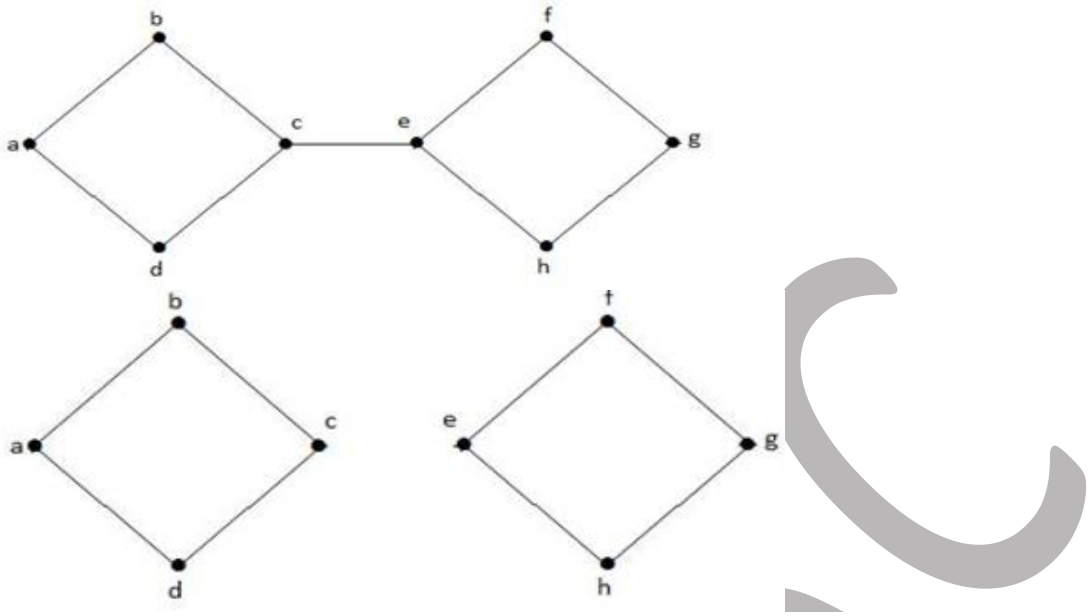
- A connected graph G may have at most (n-1) cut edges.
- Removing a cut edge may leave a graph disconnected.
- Removal of an edge may increase the number of components in a graph by at most one.
- A cut edge 'e' must not be the part of any cycle in G.
- If a cut edge exists, then a cut vertex must also exist because at least one vertex of a cut edge is a cut vertex.
- If a cut vertex exists, then the existence of any cut edge is not necessary.

Example 1



In the above graph, edge (c, e) is a cut-edge. After removing this edge from the above graph the graph will become a disconnected graph.

Example 2



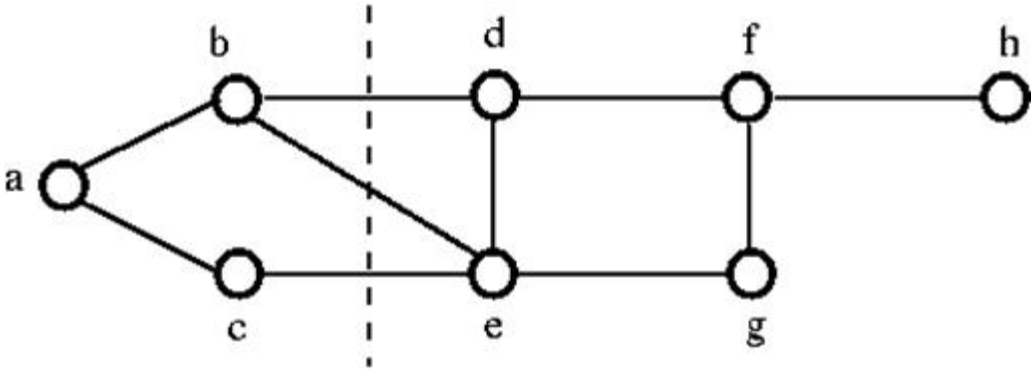
In the above graph, edge (c, e) is a cut-edge. After removing this edge from the above graph the graph will become a disconnected graph.

3. Cut Set

In a connected graph G, a cut set is a set S of edges with the following properties:

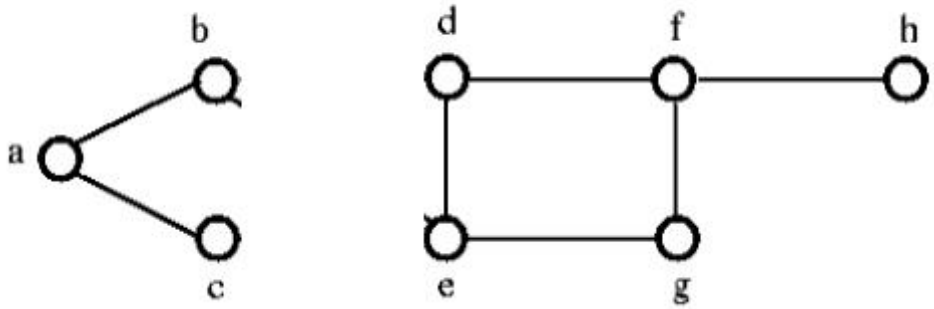
- The removal of all the edges in S disconnects G.
- The removal of some of edges (but not all) in S does not disconnect G.

Example 1



To disconnect the above graph G, we have to remove the three edges. i.e. bd, be and ce. We cannot disconnect it by removing just two of three edges. Hence, {bd, be, ce} is a cut set.

After removing the cut set from the above graph, it would look like as follows:



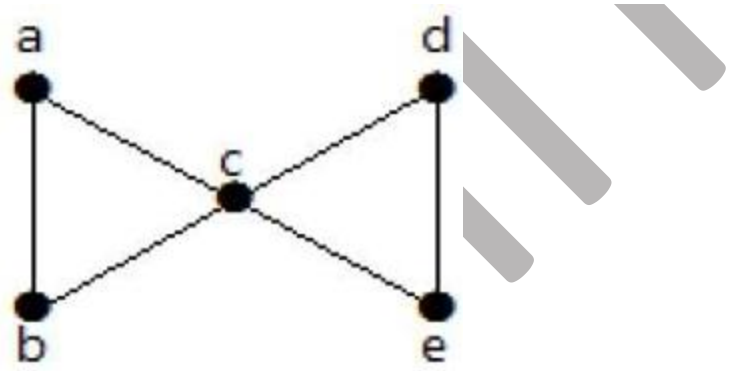
4. Edge Connectivity

The **edge connectivity** of a connected graph G is the minimum number of edges whose removal makes G disconnected. It is denoted by **λ(G)**.

When λ(G) ≥ k, then graph G is said to be **k-edge-connected**.

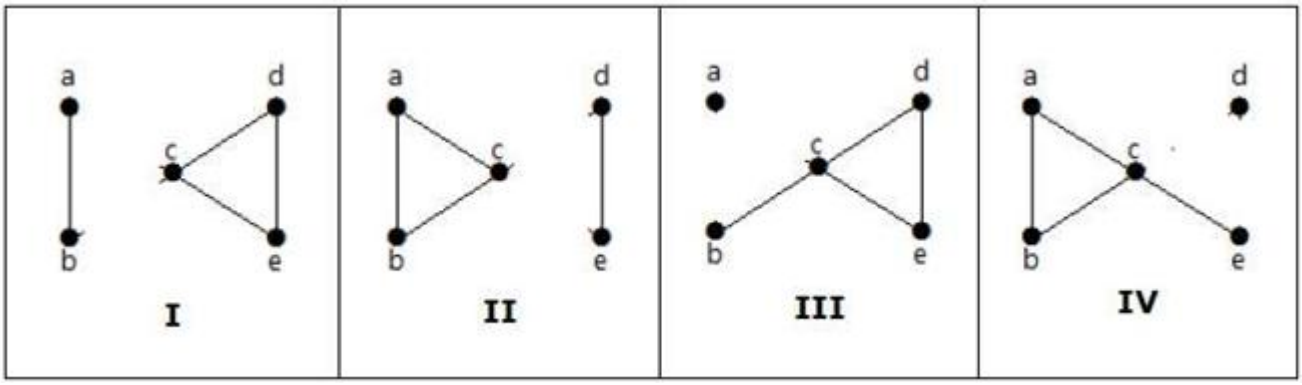
Example

Let's see an example,



From the above graph, by removing two minimum edges, the connected graph becomes disconnected graph. Hence, its edge connectivity is 2. Therefore the above graph is a **2-edge-connected graph**.

Here are the following four ways to disconnect the graph by removing two edges:



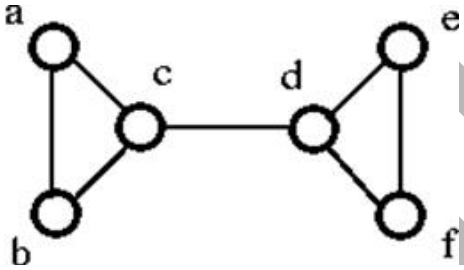
5. Vertex Connectivity

The connectivity (or vertex connectivity) of a connected graph G is the minimum number of vertices whose removal makes G disconnects or reduces to a trivial graph. It is denoted by K(G).

The graph is said to be k- connected or k-vertex connected when K(G) ≥ k. To remove a vertex we must also remove the edges incident to it.

Example

Let's see an example:



The above graph G can be disconnected by removal of the single vertex either 'c' or 'd'. Hence, its vertex connectivity is 1. Therefore, it is a 1-connected graph.

Cycles

Definition: A cycle is a closed trail in which the "first vertex = last vertex" is the only vertex that is repeated. A graph is acyclic if it does not contain a cycle. A cycle is a closed path in a graph that forms a loop. When the starting and ending point is the same in a graph that contains a set of vertices, then the cycle of the graph is formed. When there is no repetition of the vertex in a closed circuit, then the cycle is a simple cycle. The cycle graph is denoted by C_n.

- A cycle that has an even number of edges or vertices is called Even Cycle.
- A cycle that has an odd number of edges or vertices is called Odd Cycle.

In graph theory, a **cycle** in a graph is a non-empty trail in which the only repeated vertices are the first and last vertices. A **directed cycle** in a directed graph is a non-empty directed trail in which the only repeated vertices are the first and last vertices.

A graph without cycles is called an *acyclic graph*. A directed graph without directed cycles is called a *directed acyclic graph*. A connected graph without cycles is called a *tree*.

Circuit, cycle

- A **circuit** is a non-empty trail in which the first vertex is equal to the last vertex (*closed trail*).

Let $G = (V, E, \phi)$ be a graph. A circuit is a non-empty trail $(e_1, e_2, \ldots, e_n)$ with a vertex sequence $(v_1, v_2, \ldots, v_n, v_1)$.

- A **cycle** or **simple circuit** is a circuit in which the only repeated vertex is the first/last vertex.[1]
- The **length** of a circuit or cycle is the number of edges involved.

Directed circuit, cycle

- A **directed circuit** is a non-empty directed trail in which the first vertex is equal to the last vertex.[1]

Let $G = (V, E, \phi)$ be a directed graph. A directed circuit is a non-empty directed trail $(e_1, e_2, \ldots, e_n)$ with a vertex sequence $(v_1, v_2, \ldots, v_n, v_1)$.

- A **directed cycle** or **simple directed circuit** is a directed circuit in which the only repeated vertex is the first/last vertex.[1]

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

UNIT-4(DISCRETE MATHEMATICAL STRUCTURE)

SHORTEST PATH IN WEIGHTED GRAPH

LECTURE NO: -29

Shortest Path in Weighted Graph

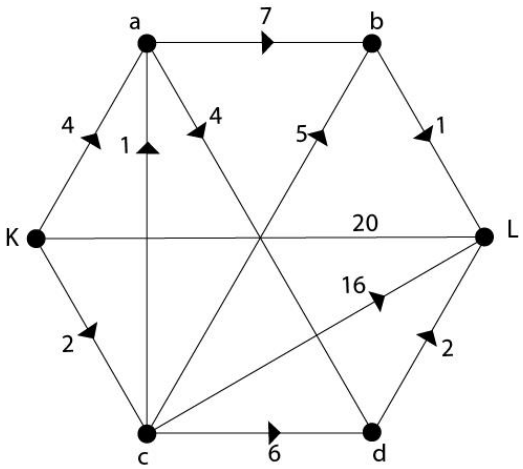
Dijkstra's Algorithm:

This algorithm maintains a set of vertices whose shortest paths from source is already known. The graph is represented by its cost adjacency matrix, where cost is the weight of the edge. In the cost adjacency matrix of the graph, all the diagonal values are zero. If there is no path from source vertex V_s to any other vertex V_i then it is represented by $+\infty$.In this algorithm, we have assumed all weights are positive.

1. Initially, there is no vertex in sets.
2. Include the source vertex V_s in S.Determine all the paths from V_s to all other vertices without going through any other vertex.
3. Now, include that vertex in S which is nearest to V_s and find the shortest paths to all the vertices through this vertex and update the values.
4. Repeat the step until n-1 vertices are not included in S if there are n vertices in the graph.

After completion of the process, we got the shortest paths to all the vertices from the source vertex.

Example: Find the shortest paths between K and L in the graph shown in fig using Dijkstra's Algorithm.



Solution:

Step1: Include the vertex K is S and determine all the direct paths from K to all other vertices without going through any other vertex.

Distance to all other vertices

S	K	a	b	c	d	L
K	0	4(K)	∞	2(K)	∞	20(K)

Step2: Include the vertex in S which is nearest to K and determine shortest paths to all vertices through this vertex and update the values. The closest vertex is c.

Distance to all other vertices

S	K	a	b	c	d	L
K	0	3(K, c)	7(K, c)	2(K)	8(K, c)	18(K, c)

Step3: The vertex which is 2nd nearest to K is 9, included in S.

Distance to all other vertices

S	K	a	b	c	d	L
K	0	3(K, c)	7(K, c)	2(K)	7(K, c, a)	18(K, c)

Step4: The vertex which is 3rd nearest to K is b, included in S.

Distance to all other vertices

S	K	a	b	c	d	L
K	0	3(K, c)	7(K, c)	2(K)	7(K, c, a)	8(K, c, b)

Step5: The vertex which is next nearest to K is d, is included in S.

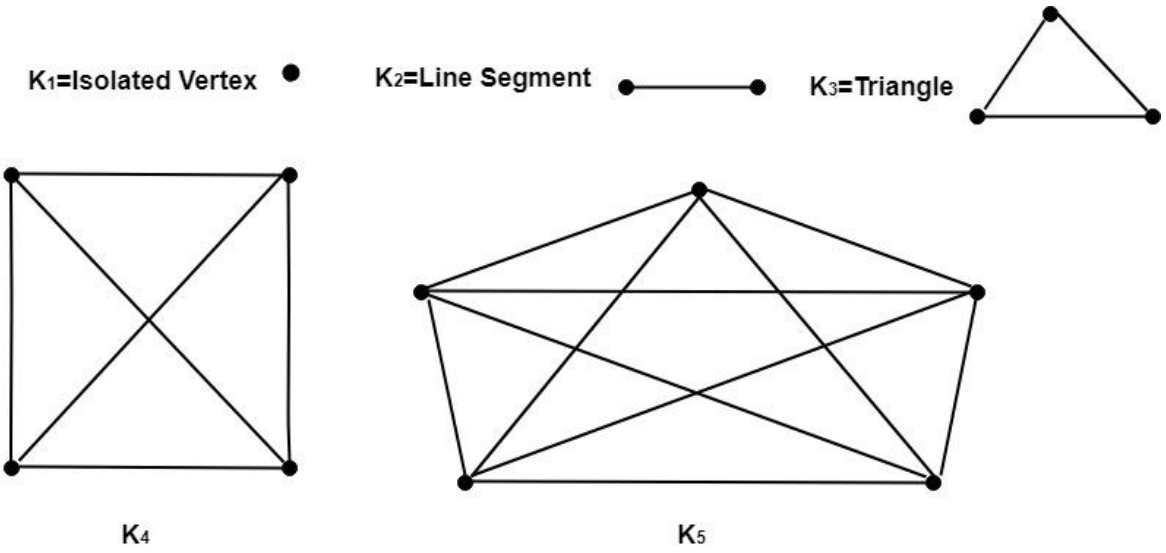
Distance to all other vertices

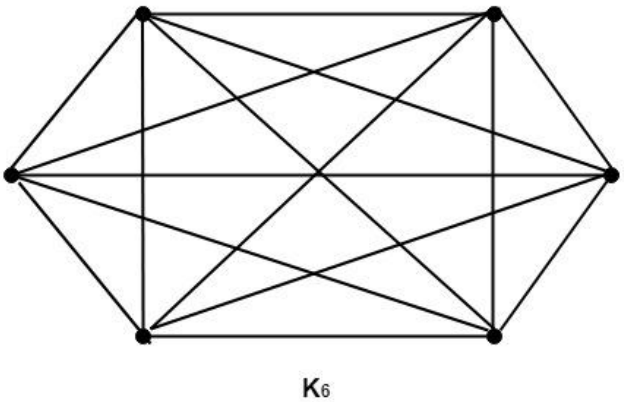
S	K	a	b	c	d	L
K(c, a, b, d)	0	3(K, c)	7(K, c)	2(K)	7(K, c, a)	8(K, c, b)

Since, n-1 vertices included in S. Hence we have found the shortest distance from K to all other vertices. Thus, the shortest distance between K and L is 8 and the shortest path is K, c, b, L.

Complete Graph

A graph G is said to be complete if every vertex in G is connected to every other vertex in G. Thus a complete graph G must be connected. The complete graph with n vertices is denoted by K_n. The Figure shows the graphs K_1 through K_6.





Regular Graph:

A graph is said to be regular or K-regular if all its vertices have the same degree K. A graph whose all vertices have degree 2 is known as a 2-regular graph. A complete graph K_n is a regular of degree n-1.

Example1: Draw regular graphs of degree 2 and 3.

Solution: The regular graphs of degree 2 and 3 are shown in fig:

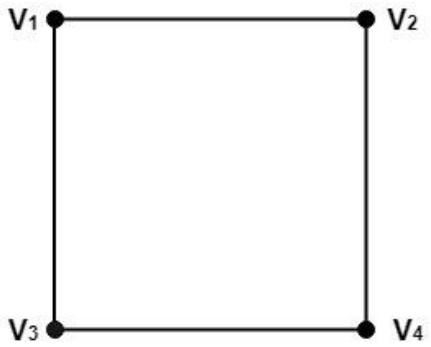


Fig:2-Regular Graph

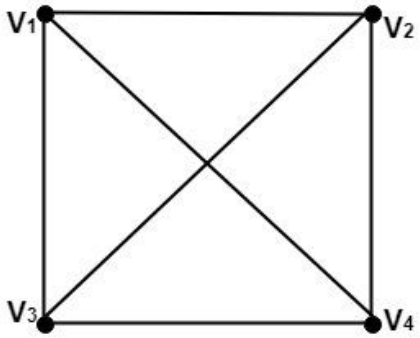
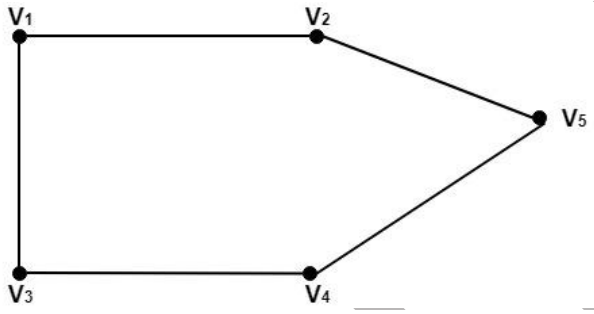


Fig:Regular Graph

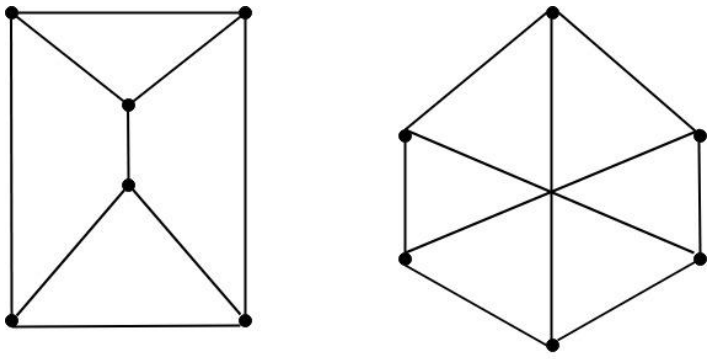
Example2: Draw a 2-regular graph of five vertices.

Solution: The 2-regular graph of five vertices is shown in fig:



Example3: Draw a 3-regular graph of five vertices.

Solution: It is not possible to draw a 3-regular graph of five vertices. The 3-regular graph must have an even number of vertices.



3-Regular Graph

Bipartite Graph:

A graph G=(V, E) is called a bipartite graph if its vertices V can be partitioned into two subsets V_1 and V_2 such that each edge of G connects a vertex of V_1 to a vertex V_2. It is denoted by K_{mn}, where m and n are the numbers of vertices in V_1 and V_2 respectively.

Example: Draw the bipartite graphs K_2, 4and K_3 ,4.Assuming any number of edges.

Solution: First draw the appropriate number of vertices on two parallel columns or rows and connect the vertices in one column or row with the vertices in other column or row. The bipartite graphs $K_{2,4}$ and $K_{3,4}$ are shown in fig respectively.

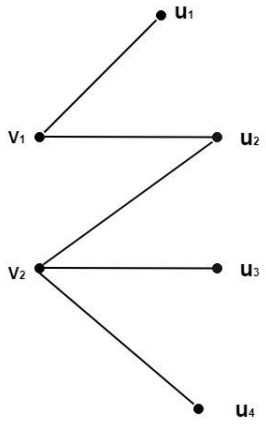


Fig:Bipartite Graph $K_{2,4}$

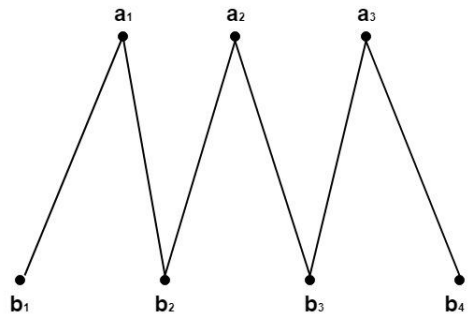


Fig:Bipartite Graph $K_{3,4}$

Complete Bipartite Graph:

A graph G = (V, E) is called a complete bipartite graph if its vertices V can be partitioned into two subsets V_1 and V_2 such that each vertex of V_1 is connected to each vertex of V_2. The number of edges in a complete bipartite graph is m.n as each of the m vertices is connected to each of the n vertices.

Example: Draw the complete bipartite graphs $K_{3,4}$ and $K_{1,5}$.

Solution: First draw the appropriate number of vertices in two parallel columns or rows and connect the vertices in the first column or row with all the vertices in the second column or row. The graphs $K_{3,4}$ and $K_{1,5}$ are shown in fig:

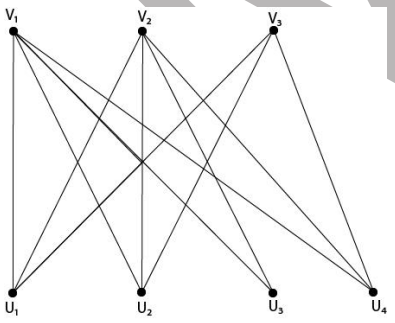


Fig: $K_{3,4}$

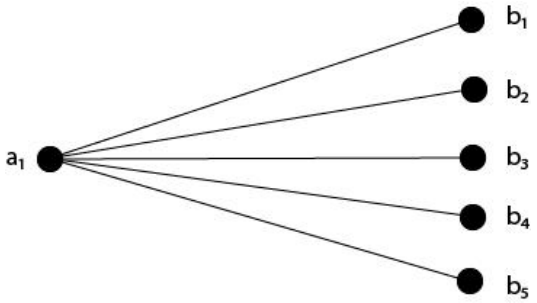


Fig: $K_{1,5}$

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

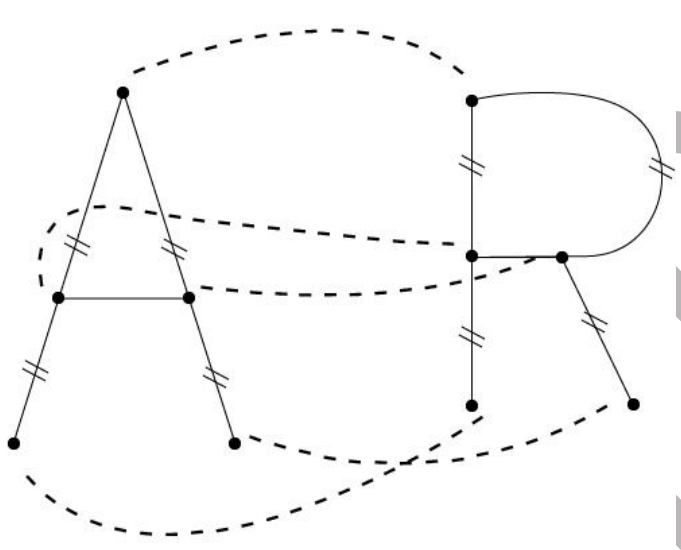
UNIT-4 (DISCRETE MATHEMATICAL STRUCTURE)

ISOMORPHISM AND HOMOMORPHISM OF GRAPHS

LECTURE NO: -30

Isomorphic Graphs

Consider a graph G(V, E) and G* (V*,E*) are said to be isomorphic if there exists one to one correspondence i.e. f:V→V* such that {u, v} is an edge of G if and only if {f(u), f(v)} is an edge of G*.



Number of vertices of graph (a) must be equal to graph (b), i.e., one to one correspondence some goes for edges.

Homeomorphic Graphs:

Two graphs G and G* are said to homeomorphic if they can be obtained from the same graph or isomorphic graphs by this method. The graphs (a) and (b) are not isomorphic, but they are homeomorphic since they can be obtained from the graph (c) by adding appropriate vertices.

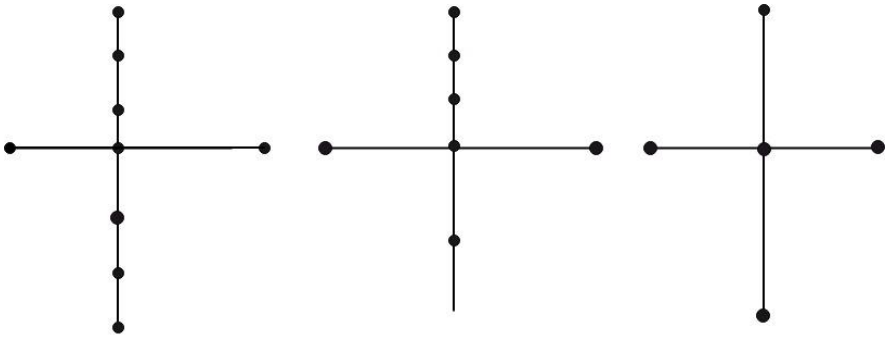


Fig:a Fig:b Fig:c

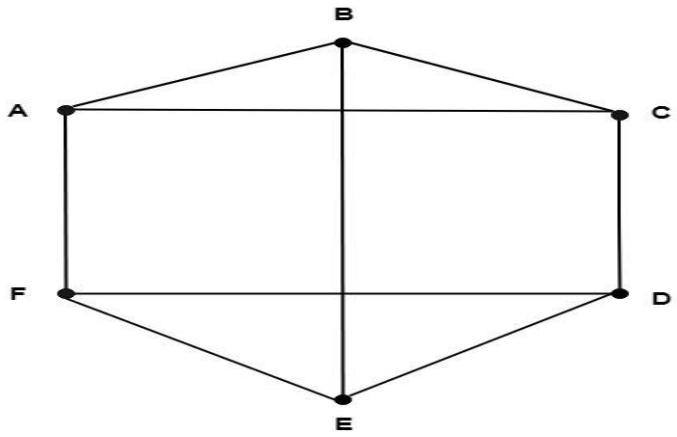
Notes By: -Sunil kumar Sharma (Assistant Professor-Maths Department)

Subgraph:

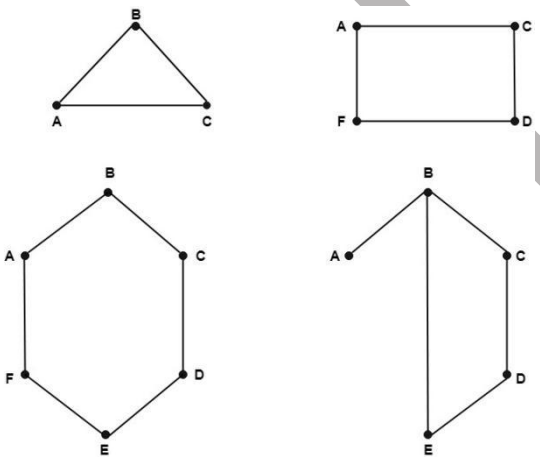
A subgraph of a graph G=(V, E) is a graph G'=(V',E') in which V'⊆V and E'⊆E and each edge of G' have the same end vertices in G' as in graph G.

Note: A single vertex is a subgraph.

Example: Consider the graph G shown in fig. Show the different subgraph of this graph.



Solution: The following are all subgraphs of the above graph as shown in fig:



Spanning Subgraph:

A graph G_1 is called a spanning subgraph of G if G_1 contains all the vertices of G.

Example: The following fig is the spanning subgraph of the graph shown in Fig:

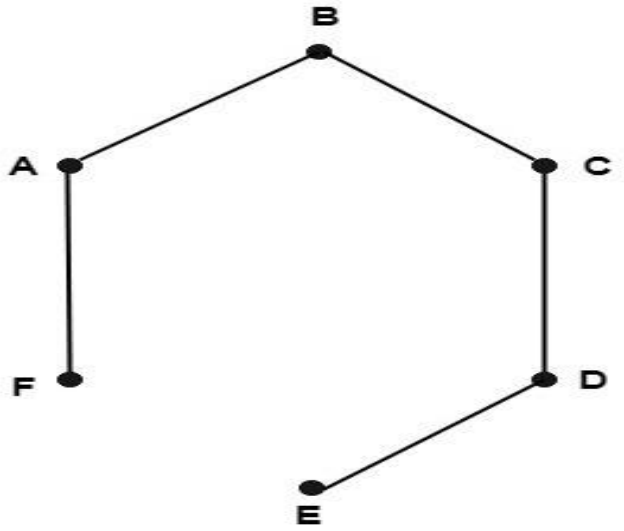


Fig:Spanning Subgraph

Isomorphism

Two graphs G_1 and G_2 are said to be isomorphic if –

- Their number of components (vertices and edges) are same.
- Their edge connectivity is retained.

Note − In short, out of the two isomorphic graphs, one is a tweaked version of the other. An unlabelled graph also can be thought of as an isomorphic graph.

If $G_1 = G_2$ then −

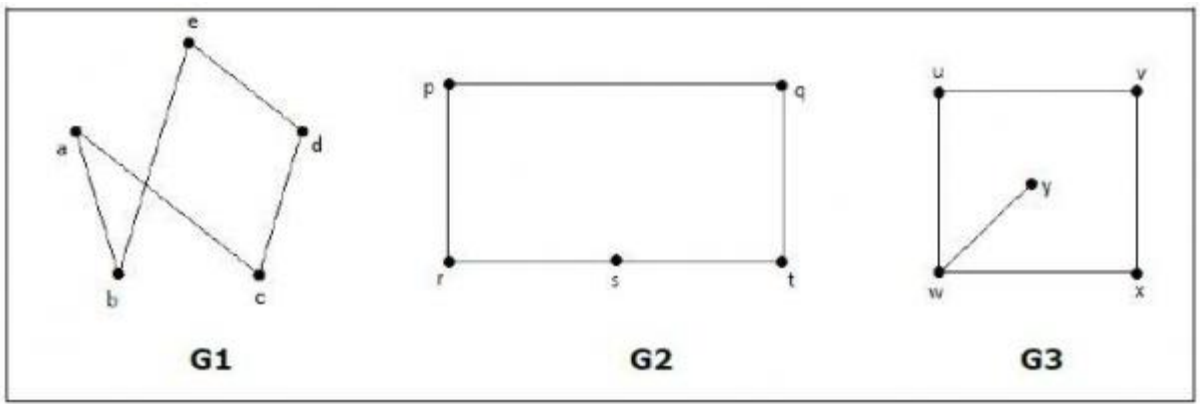
- $|V(G_1)| = |V(G_2)|$
- $|E(G_1)| = |E(G_2)|$
- Degree sequences of G_1 and G_2 are same.
- If the vertices $\{V_1, V_2, .. V_k\}$ form a cycle of length K in G_1, then the vertices $\{f(V_1), f(V_2),… f(V_k)\}$ should form a cycle of length K in G_2.

All the above conditions are necessary for the graphs G_1 and G_2 to be isomorphic, but not sufficient to prove that the graphs are isomorphic.

- $(G_1 = G_2)$ if and only if $(G_1 - = G_2 -)$ where G_1 and G_2 are simple graphs.
- $(G_1 = G_2)$ if the adjacency matrices of G_1 and G_2 are same.
- $(G_1 = G_2)$ if and only if the corresponding subgraphs of G_1 and G_2 (obtained by deleting some vertices in G_1 and their images in graph G_2) are isomorphic.

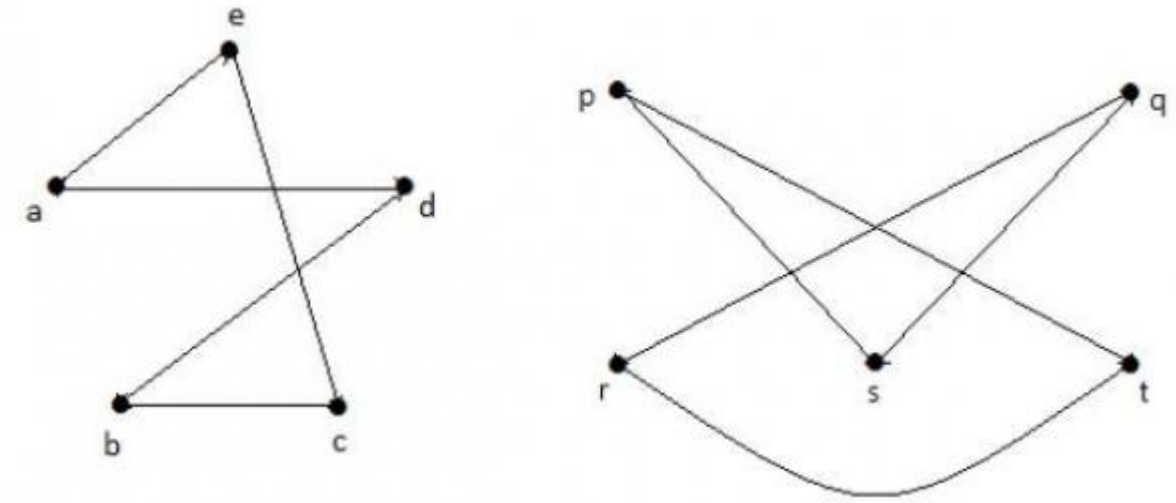
Example

Which of the following graphs are isomorphic?



In the graph G_3, vertex 'w' has only degree 3, whereas all the other graph vertices has degree 2. Hence G_3 not isomorphic to G_1 or G_2.

Taking complements of G_1 and G_2, you have −



Here, (G_1− = G_2−), hence (G_1 = G_2).

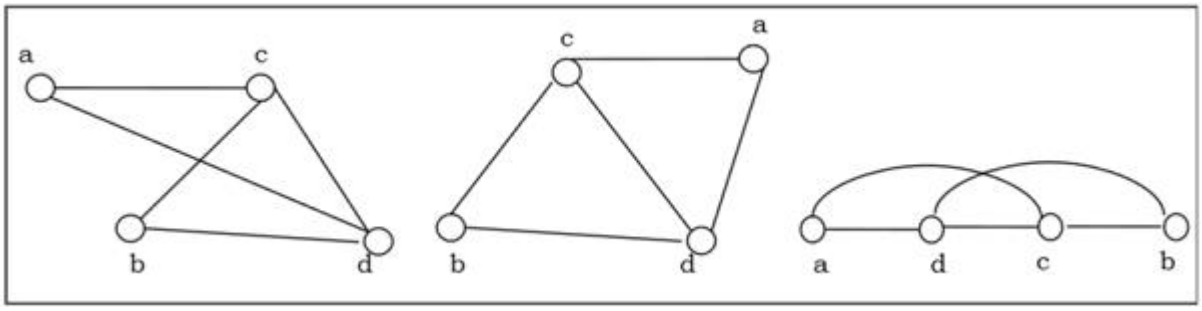
If two graphs G and H contain the same number of vertices connected in the same way, they are called isomorphic graphs (denoted by G ≅ H).

It is easier to check non-isomorphism than isomorphism. If any of these following conditions occurs, then two graphs are non-isomorphic −

- The number of connected components are different
- Vertex-set cardinalities are different
- Edge-set cardinalities are different
- Degree sequences are different

Example

The following graphs are isomorphic −



Homomorphism

A homomorphism from a graph ***G*** to a graph ***H*** is a mapping (May not be a bijective mapping) h: G → H such that – (x, y) ∈ E(G) → (h(x), h(y)) ∈ E(H). It maps adjacent vertices of graph G to the adjacent vertices of the graph ***H***.

Properties of Homomorphisms

- A homomorphism is an isomorphism if it is a bijective mapping.
- Homomorphism always preserves edges and connectedness of a graph.
- The compositions of homomorphisms are also homomorphisms.
- To find out if there exists any homomorphic graph of another graph is a NPcomplete problem.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

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UNIT-4 (DISCRETE MATHEMATICAL STRUCTURE)

EULARIAN AND HAMILTONIAN PATHS, CIRCUITS AND GRAPHS

LECTURE NO: 31

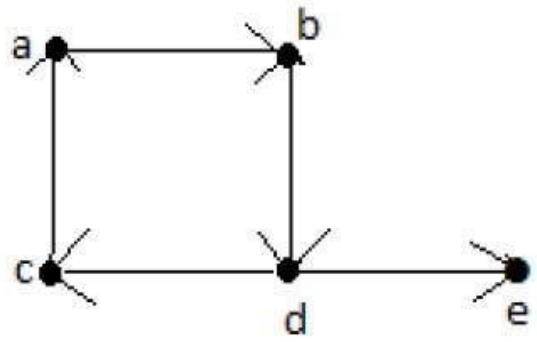
Eularian and Hamiltonian Paths, Circuits and Graphs

A graph is traversable if you can draw a path between all the vertices without retracing the same path. Based on this path, there are some categories like Euler's path and Euler's circuit which are described in this chapter.

Euler's Path

An Euler's path contains each edge of 'G' exactly once and each vertex of 'G' at least once. A connected graph G is said to be traversable if it contains an Euler's path.

Example

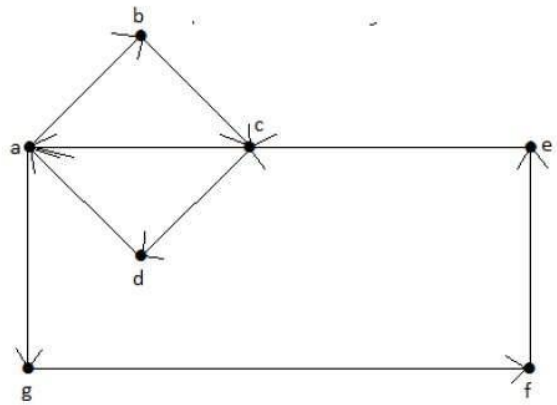


Euler's Path = d-c-a-b-d-e.

Euler's Circuit

In an Euler's path, if the starting vertex is same as its ending vertex, then it is called an Euler's circuit.

Example



Notes By: -Sunil kumar Sharma (Assistant Professor-Maths Department)

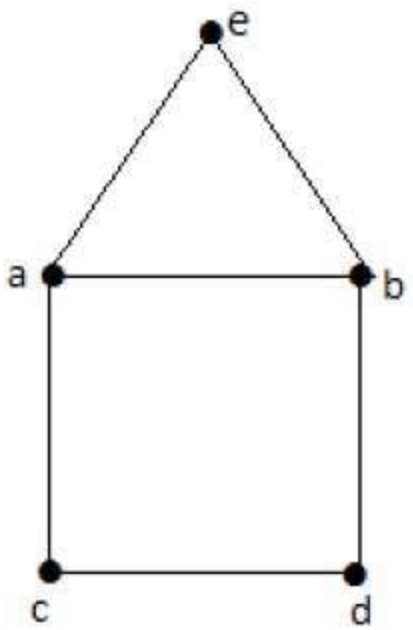
Euler's Path = a-b-c-d-a-g-f-e-c-a.

Euler's Circuit Theorem

A connected graph 'G' is traversable if and only if the number of vertices with odd degree in G is exactly 2 or 0. A connected graph G can contain an Euler's path, but not an Euler's circuit, if it has exactly two vertices with an odd degree.

Note − This Euler path begins with a vertex of odd degree and ends with the other vertex of odd degree.

Example



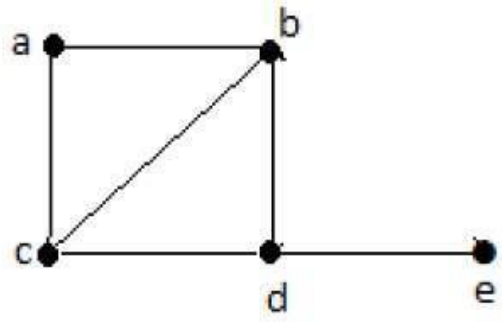
Euler's Path − b-e-a-b-d-c-a is not an Euler's circuit, but it is an Euler's path. Clearly it has exactly 2 odd degree vertices.

Note − In a connected graph G, if the number of vertices with odd degree = 0, then Euler's circuit exists.

Hamiltonian Path

A connected graph is said to be Hamiltonian if it contains each vertex of G exactly once. Such a path is called a **Hamiltonian path**.

Example



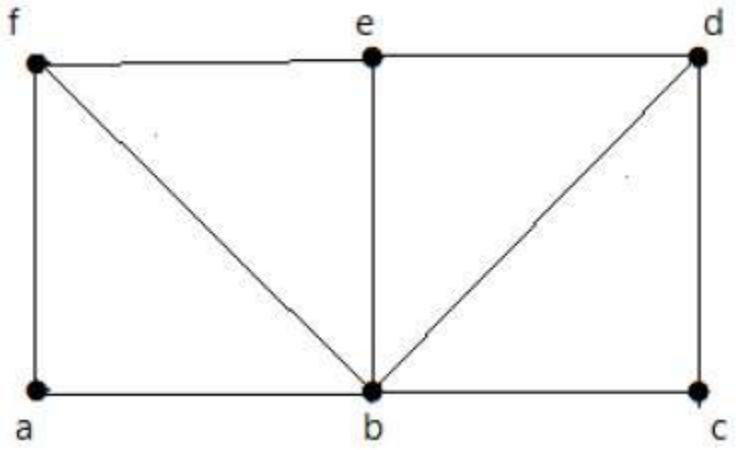
Hamiltonian Path − e-d-b-a-c.

Note −

- Euler's circuit contains each edge of the graph exactly once.
- In a Hamiltonian cycle, some edges of the graph can be skipped.

Example

Take a look at the following graph −



For the graph shown above −

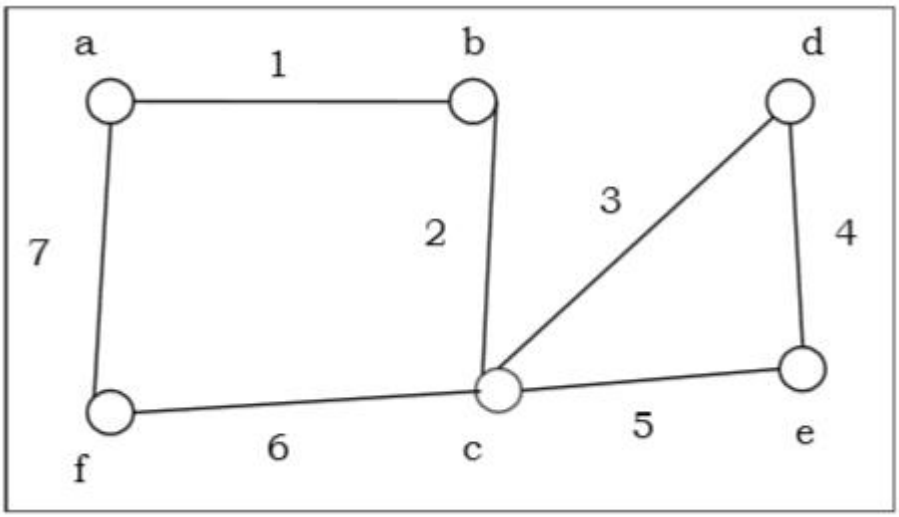
- Euler path exists – false
- Euler circuit exists – false
- Hamiltonian cycle exists – true
- Hamiltonian path exists – true

G has four vertices with odd degree, hence it is not traversable. By skipping the internal edges, the graph has a Hamiltonian cycle passing through all the vertices.

Euler Graph - A connected graph G is called an Euler graph, if there is a closed trail which includes every edge of the graph G.

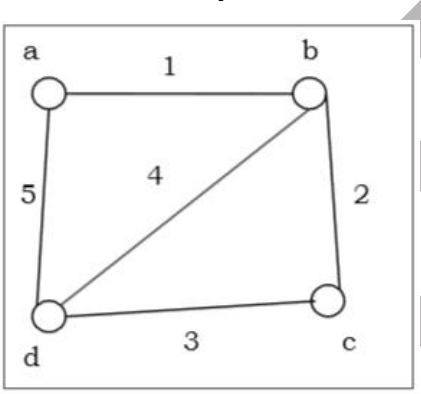
Euler Path - An Euler path is a path that uses every edge of a graph exactly once. An Euler path starts and ends at different vertices.

Euler Circuit - An Euler circuit is a circuit that uses every edge of a graph exactly once. An Euler circuit always starts and ends at the same vertex. A connected graph G is an Euler graph if and only if all vertices of G are of even degree, and a connected graph G is Eulerian if and only if its edge set can be decomposed into cycles.



The above graph is an Euler graph as a 1 b 2 c 3 d 4 e 5 c 6 f 7 g covers all the edges of the graph.

Non-Euler Graph

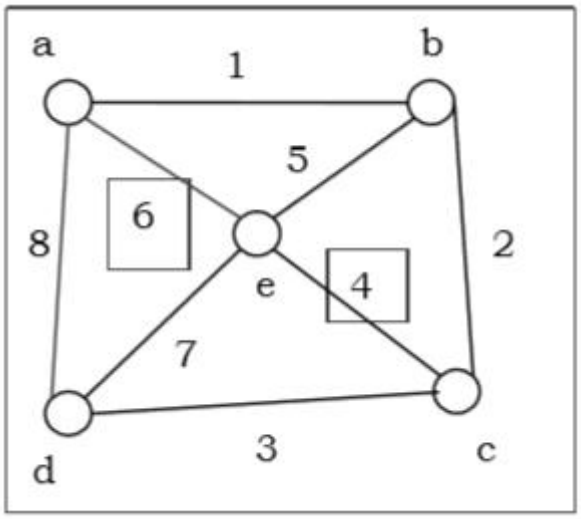


Here degree of vertex b and d is 3, an odd degree and violating the euler graph condition.

Hamiltonian graph - A connected graph G is called Hamiltonian graph if there is a cycle which includes every vertex of G and the cycle is called Hamiltonian cycle. Hamiltonian walk in graph G is a walk that passes through each vertex exactly once.

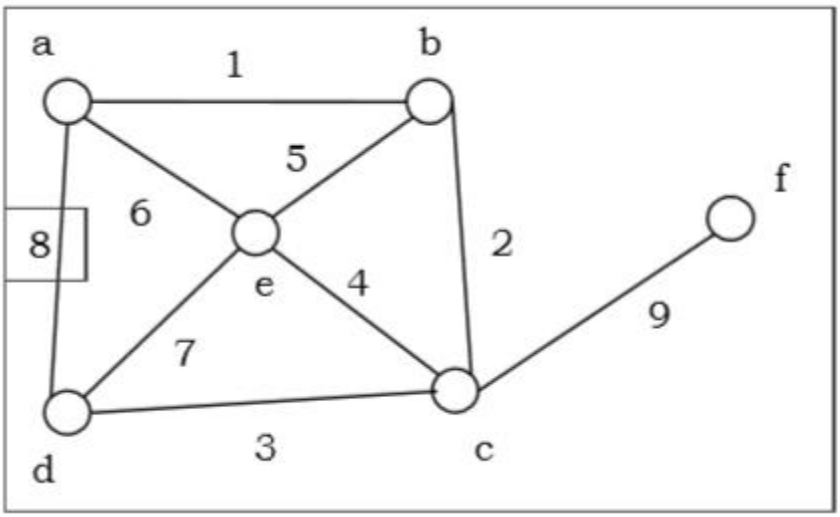
Dirac's Theorem - If G is a simple graph with n vertices, where n ≥ 3 If deg(v) ≥ {n}/{2} for each vertex v, then the graph G is Hamiltonian graph.

Ore's Theorem - If G is a simple graph with n vertices, where n ≥ 2 if deg(x) + deg(y) ≥ n for each pair of non-adjacent vertices x and y, then the graph G is Hamiltonian graph.



In above example, sum of degree of a and c vertices is 6 and is greater than total vertices, 5 using Ore's theorem, it is an Hamiltonian Graph.

Non-Hamiltonian Graph



In above example, sum of degree of a and f vertices is 4 and is less than total vertices, 4 using Ore's theorem, it is not an Hamiltonian Graph.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

UNIT-4 (DISCRETE MATHEMATICAL STRUCTURE)

GRAPH COLORING AND CHROMATIC NUMBER

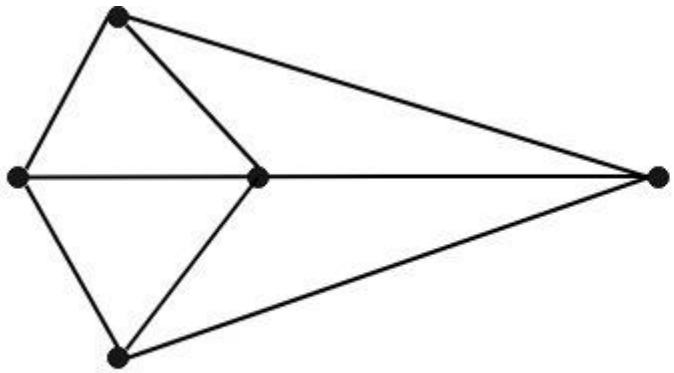
LECTURE NO: 32

Graph Coloring

Suppose that G= (V,E) is a graph with no multiple edges. A vertex coloring of G is an assignment of colors to the vertices of G such that adjacent vertices have different colors. A graph G is M-Colorable if there exists a coloring of G which uses M-Colors.

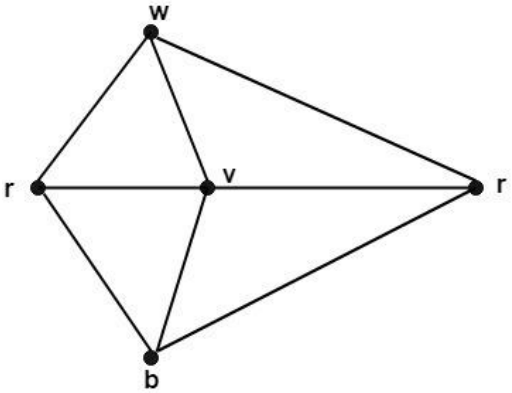
Proper Coloring: A coloring is proper if any two adjacent vertices u and v have different colors otherwise it is called improper coloring.

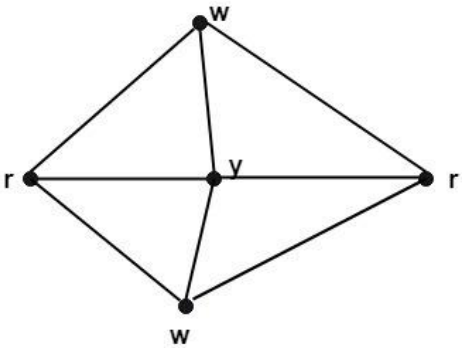
Example: Consider the following graph and color C={r, w, b, y}.Color the graph properly using all colors or fewer colors.



The graph shown in fig is a minimum 3-colorable, hence x(G)=3

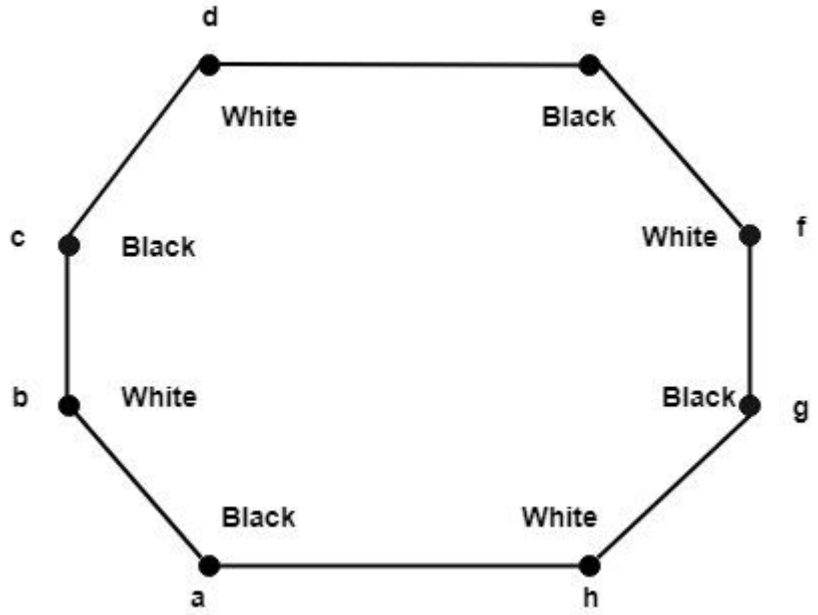
Solution: Fig shows the graph properly colored with all the four colors.





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Fig shows the graph properly colored with three colors.



Method to Color a Graph

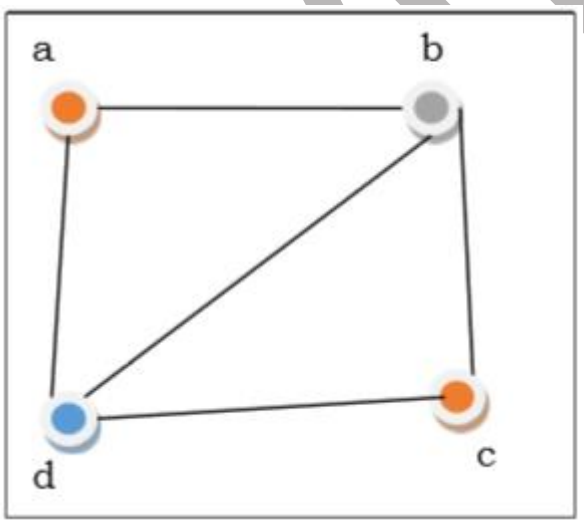
The steps required to color a graph G with n number of vertices are as follows −

Step 1 − Arrange the vertices of the graph in some order.

Step 2 − Choose the first vertex and color it with the first color.

Step 3 − Choose the next vertex and color it with the lowest numbered color that has not been colored on any vertices adjacent to it. If all the adjacent vertices are colored with this color, assign a new color to it. Repeat this step until all the vertices are colored.

Example



In the above figure, at first vertex a is colored red. As the adjacent vertices of vertex a are again adjacent, vertex b and vertex d are colored with different color, green and blue respectively. Then vertex c is colored as red as no adjacent vertex of c is colored red. Hence, we could color the graph by 3 colors. Hence, the chromatic number of the graph is 3.

Chromatic number of G: The minimum number of colors needed to produce a proper coloring of a graph G is called the chromatic number of G and is denoted by x(G).

Example: The chromatic number of K_n is n.

Solution: A coloring of K_n can be constructed using n colours by assigning different colors to each vertex. No two vertices can be assigned the same colors, since every two vertices of this graph are adjacent. Hence the chromatic number of K_n=n.

Applications of Graph Coloring

Some applications of graph coloring include:

- Register Allocation
- Map Coloring
- Bipartite Graph Checking
- Mobile Radio Frequency Assignment
- Making a time table, etc.

State and prove Handshaking Theorem.

Handshaking Theorem: The sum of degrees of all the vertices in a graph G is equal to twice the number of edges in the graph.

Mathematically it can be stated as:

$$\sum_{v \in V} deg(v) = 2e$$

Proof: Let G = (V, E) be a graph where V = $\{v_1, v_2, \ldots\}$ be the set of vertices and E = $\{e_1, e_2 \ldots\}$ be the set of edges. We know that every edge lies between two vertices so it provides degree one to each vertex. Hence each edge contributes degree two for the graph. So the sum of degrees of all vertices is equal to twice the number of edges in G.

Hence, $\sum_{v \in V} deg(v) = 2e$

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.

UNIT-4 (DISCRETE MATHEMATICAL STRUCTURE)

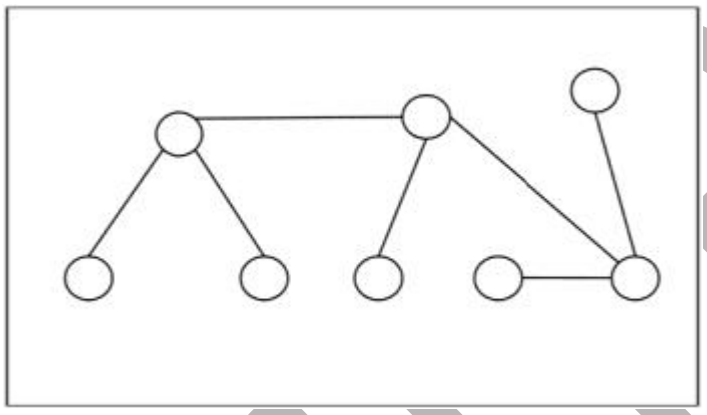
TREE AND ITS PROPERTIES

LECTURE NO: 33

Tree and its Properties

Definition − A Tree is a connected acyclic undirected graph. There is a unique path between every pair of vertices in G. A tree with N number of vertices contains (N−1) number of edges. The vertex which is of 0 degree is called root of the tree. The vertex which is of 1 degree is called leaf node of the tree and the degree of an internal node is at least 2.

Example − The following is an example of a tree −



Centers and Bi-Centers of a Tree

The center of a tree is a vertex with minimal eccentricity. The eccentricity of a vertex X in a tree G is the maximum distance between the vertex X and any other vertex of the tree. The maximum eccentricity is the tree diameter. If a tree has only one center, it is called Central Tree and if a tree has only more than one centers, it is called Bi-central Tree. Every tree is either central or bi-central.

Algorithm to find centers and bi-centers of a tree

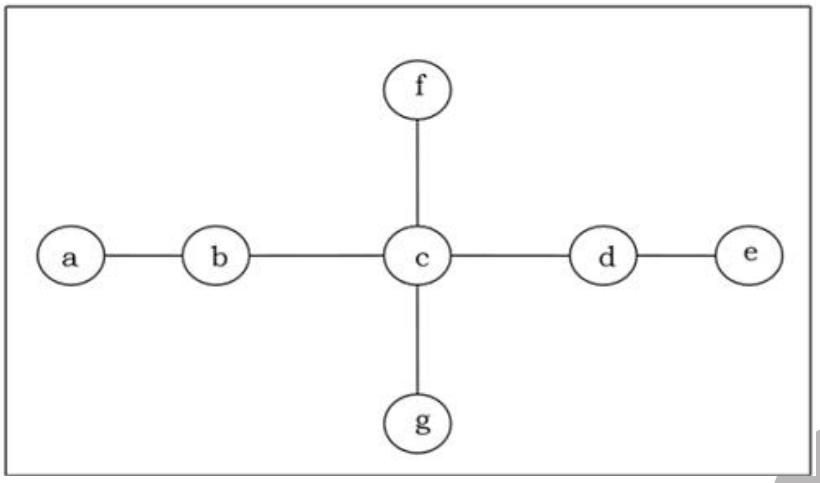
Step 1 − Remove all the vertices of degree 1 from the given tree and also remove their incident edges.

Step 2 − Repeat step 1 until either a single vertex or two vertices joined by an edge is left. If a single vertex is left then it is the center of the tree and if two vertices joined by an edge is left then it is the bi-center of the tree.

Problem 1

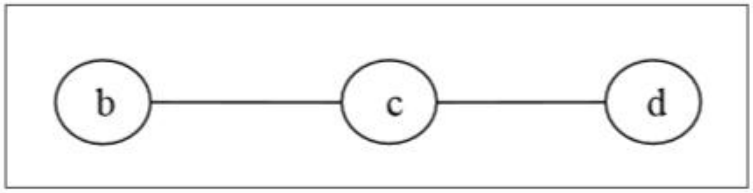
Find out the center/bi-center of the following tree −

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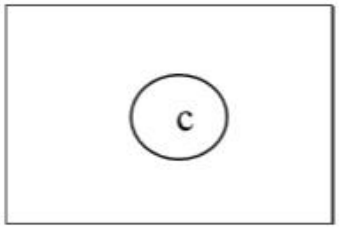


Solution

At first, we will remove all vertices of degree 1 and also remove their incident edges and get the following tree −



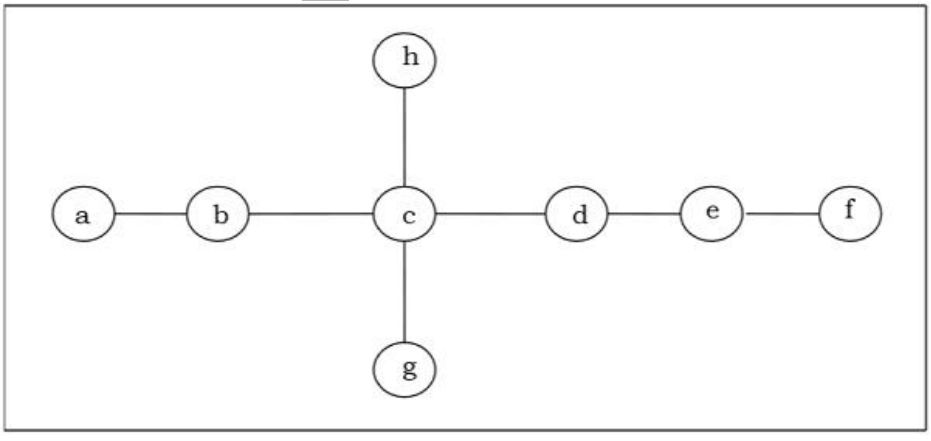
Again, we will remove all vertices of degree 1 and also remove their incident edges and get the following tree −



Finally we got a single vertex ‘c’ and we stop the algorithm. As there is single vertex, this tree has one center ‘c’ and the tree is a central tree.

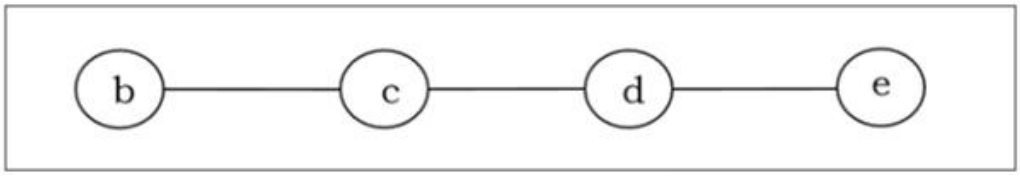
Problem 2

Find out the center/bi-center of the following tree −

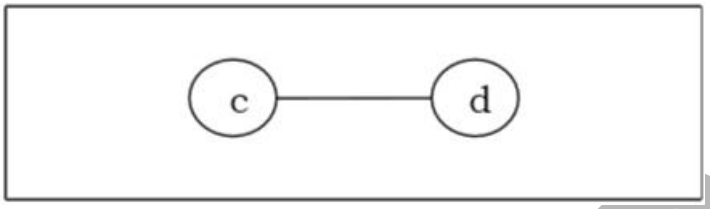


Solution

At first, we will remove all vertices of degree 1 and also remove their incident edges and get the following tree −



Again, we will remove all vertices of degree 1 and also remove their incident edges and get the following tree −

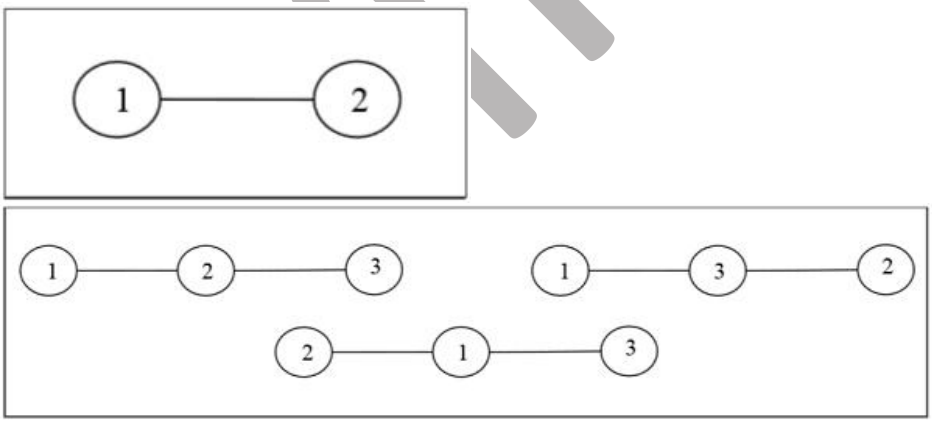


Finally, we got two vertices ‘c’ and ‘d’ left, hence we stop the algorithm. As two vertices joined by an edge is left, this tree has bi-center ‘cd’ and the tree is bi-central.

Labeled Trees

Definition − A labeled tree is a tree the vertices of which are assigned unique numbers from 1 to n. We can count such trees for small values of n by hand so as to conjecture a general formula. The number of labeled trees of n number of vertices is n^{n-2}. Two labeled trees are isomorphic if their graphs are isomorphic and the corresponding points of the two trees have the same labels.

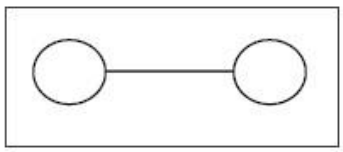
Example



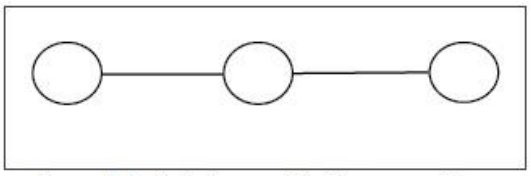
Unlabeled Trees

Definition − An unlabeled tree is a tree the vertices of which are not assigned any numbers. The number of labeled trees of n number of vertices is (2n)!/(n+1)!n! (nth Catalan number)

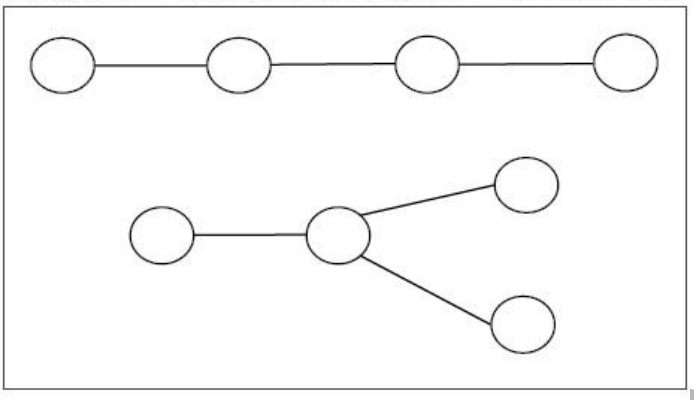
Example



An unlabeled tree with two vertices



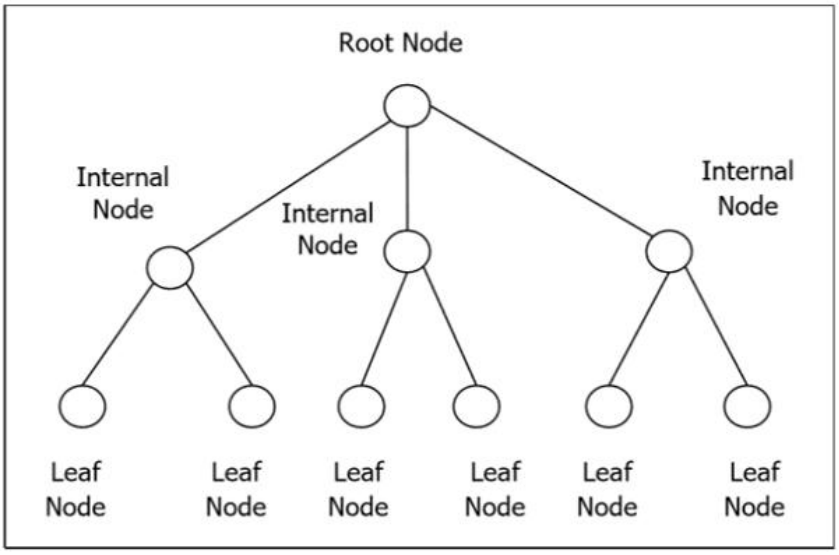
An unlabeled tree with three vertices



Two possible unlabeled trees with four vertices

Rooted Tree

A rooted tree G is a connected acyclic graph with a special node that is called the root of the tree and every edge directly or indirectly originates from the root. An ordered rooted tree is a rooted tree where the children of each internal vertex are ordered. If every internal vertex of a rooted tree has not more than m children, it is called an m-ary tree. If every internal vertex of a rooted tree has exactly m children, it is called a full m-ary tree. If m=2$\diamondsuit$=2, the rooted tree is called a binary tree.



Binary Search Tree

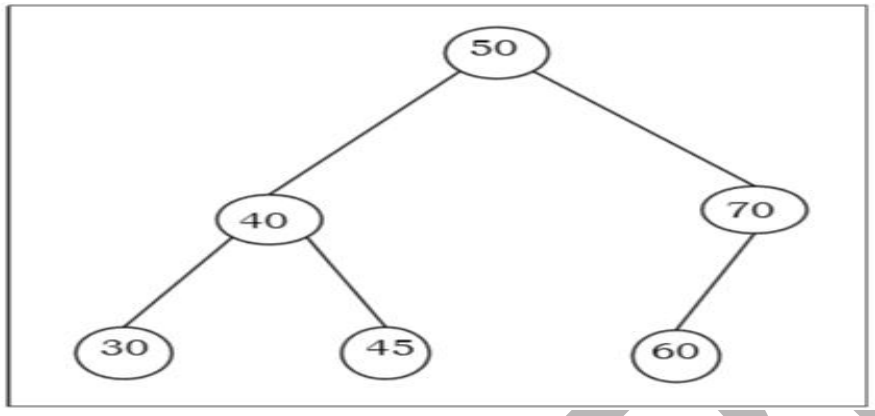
Binary Search tree is a binary tree which satisfies the following property −

- X in left sub-tree of vertex V,Value(X)≤Value(V)

- Y in right sub-tree of vertex V,Value(Y)≥Value(V)

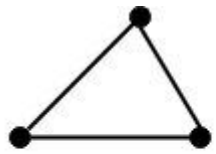
So, the value of all the vertices of the left sub-tree of an internal node V are less than or equal to V and the value of all the vertices of the right sub-tree of the internal node V are greater than or equal to V. The number of links from the root node to the deepest node is the height of the Binary Search Tree.

Example

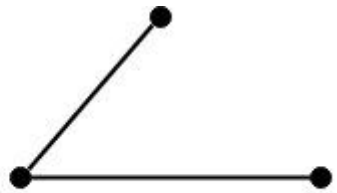


Spanning Tree

A subgraph T of a connected graph G is called spanning tree of G if T is a tree and T include all vertices of G.



Cycle Graph



Subgraph/Spanning Tree

Minimum Spanning Tree:

Suppose G is a connected weight graph i.e., each edge of G is assigned a non-negative number called the weight of edge, then any spanning tree T of G is assigned a total weight obtained by adding the weight of the edge in T.

A minimum spanning tree of G is a tree whose total weight is as small as possible.

Kruskal's Algorithm to find a minimum spanning tree: This algorithm finds the minimum spanning tree T of the given connected weighted graph G.

References:

1. Kenneth H. Rosen. Discrete Mathematics and its applications. Tata McGraw Hill.

2. Kolman, Busby and Ross, Distrete Mathematics Structures, PHI.